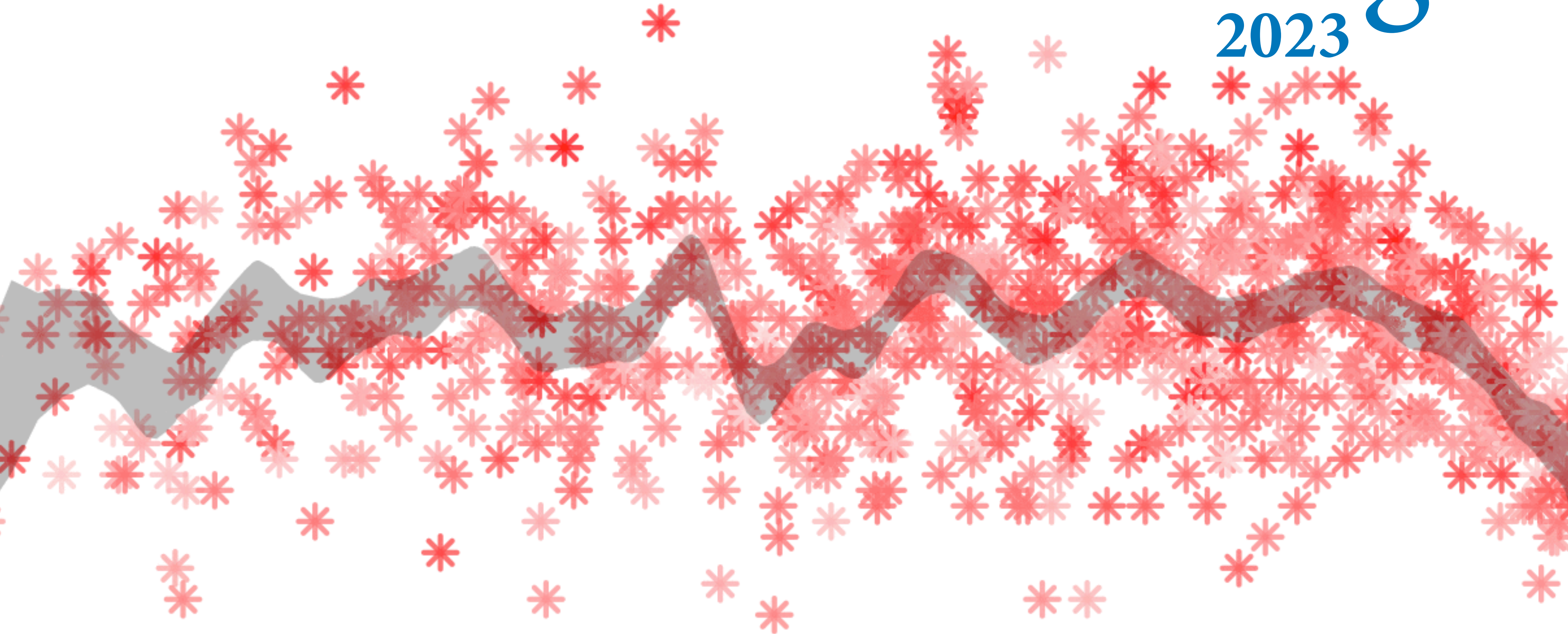


Statistical Rethinking

2023



3. Geocentric Models

Watch Statistical Rethinking 2023 - 03 - Geocentric Models on Youtube until minute 22

<https://www.youtube.com/watch?v=tNOu-SEacNU&list=PLDcUM9US4XdPz-KxHM4XHt7uUVGWWVSus>

Owl-drawing workflow

- (1) State a clear **question**
- (2) Sketch your causal **assumptions**
- (3) Use the sketch to define a **generative** model
- (4) Use generative model to build **estimator**
- (5) Profit



Linear Regression

Drawing the Owl

(1) Question/goal/estimand

(2) Scientific model

(3) Statistical model(s)

(4) Validate model

(5) Analyze data



Life Histories of the
DOBE !KUNG

FOOD, FATNESS, AND WELL-BEING OVER THE LIFE-SPAN

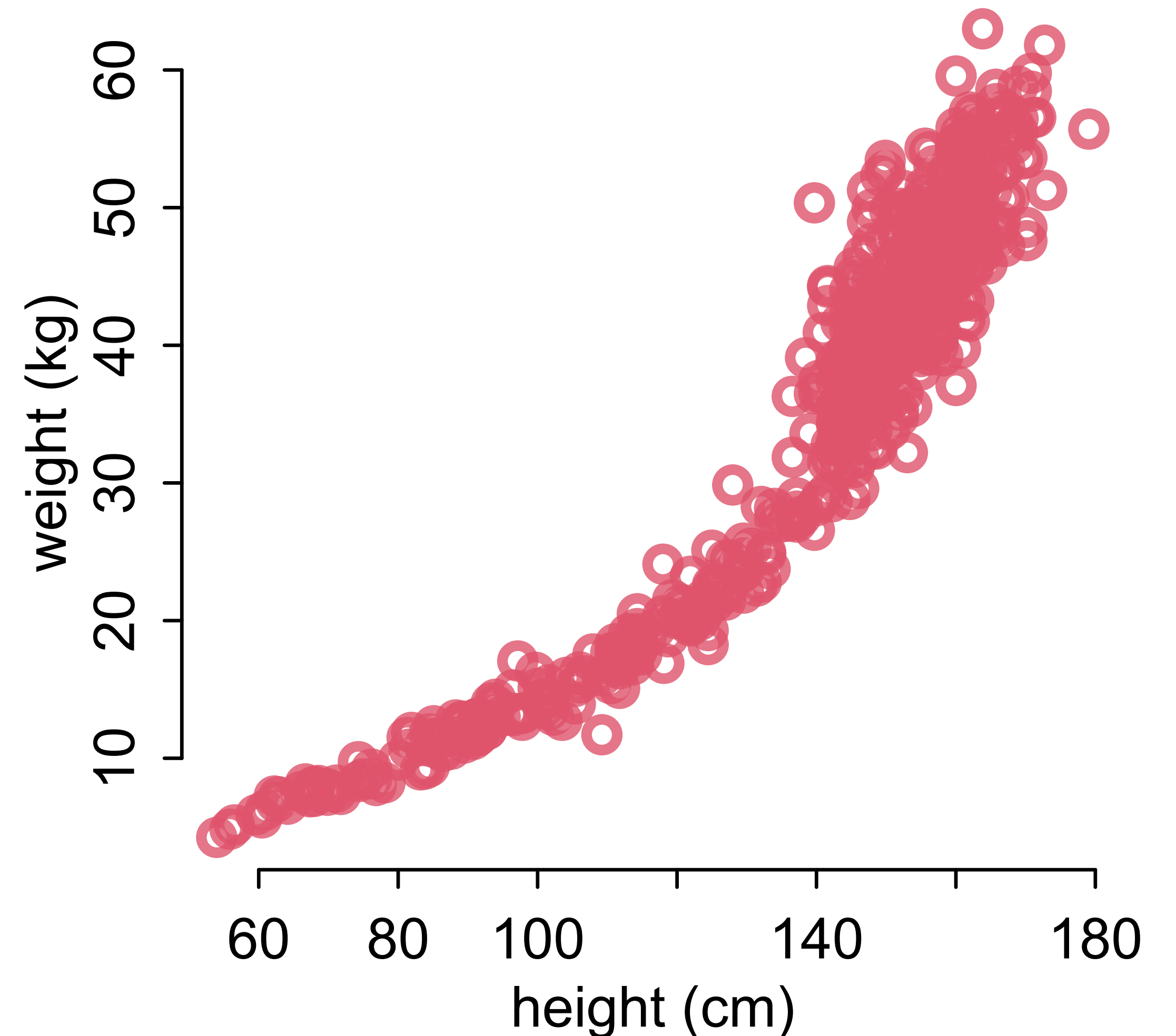
NANCY HOWELL

Linear Regression

```
library(rethinking)  
data(Howell1)
```

Drawing the Owl

- (1) Question/goal/estimand
- (2) Scientific model
- (3) Statistical model(s)
- (4) Validate model
- (5) Analyze data



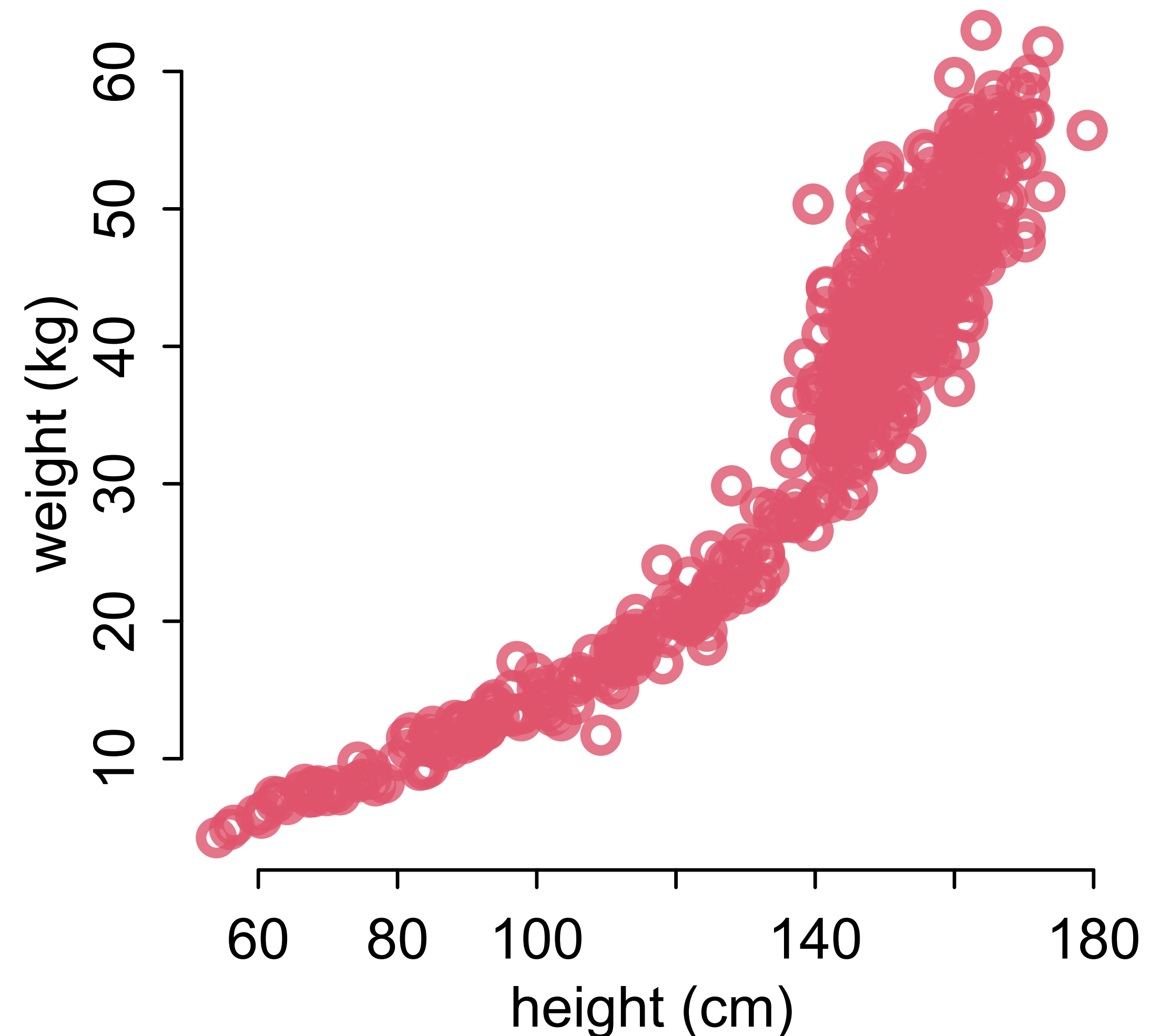
Linear Regression

```
library(rethinking)  
data(Howell1)
```

Drawing the Owl

(1) Question/goal/estimand

Describe association between
weight and **height**



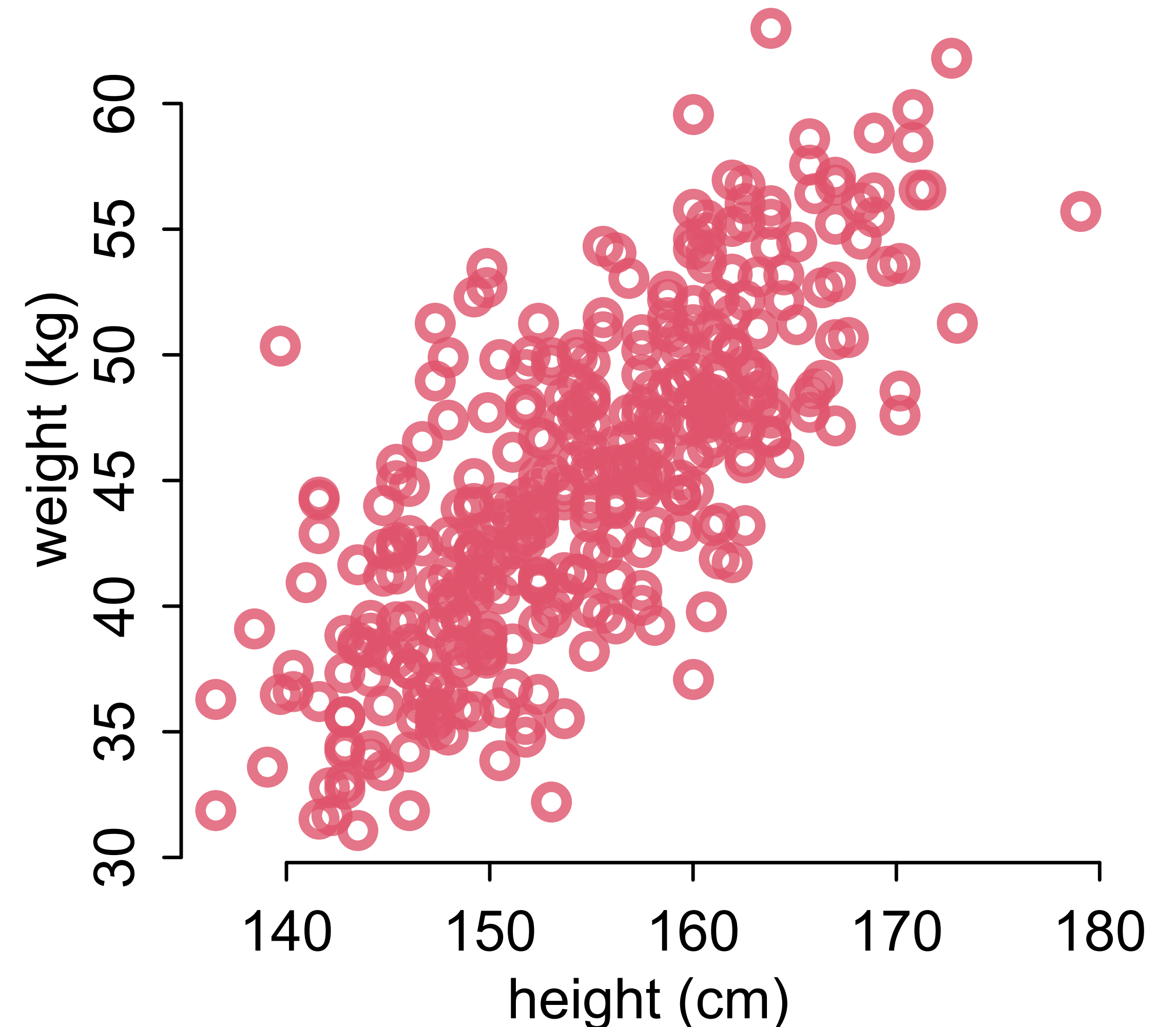
Linear Regression

```
data(Howell1)  
d <- Howell1[Howell1$age>=18,]
```

Drawing the Owl

(1) Question/goal/estimand

Describe association between
ADULT **weight** and **height**



Linear Regression

(2) Scientific model

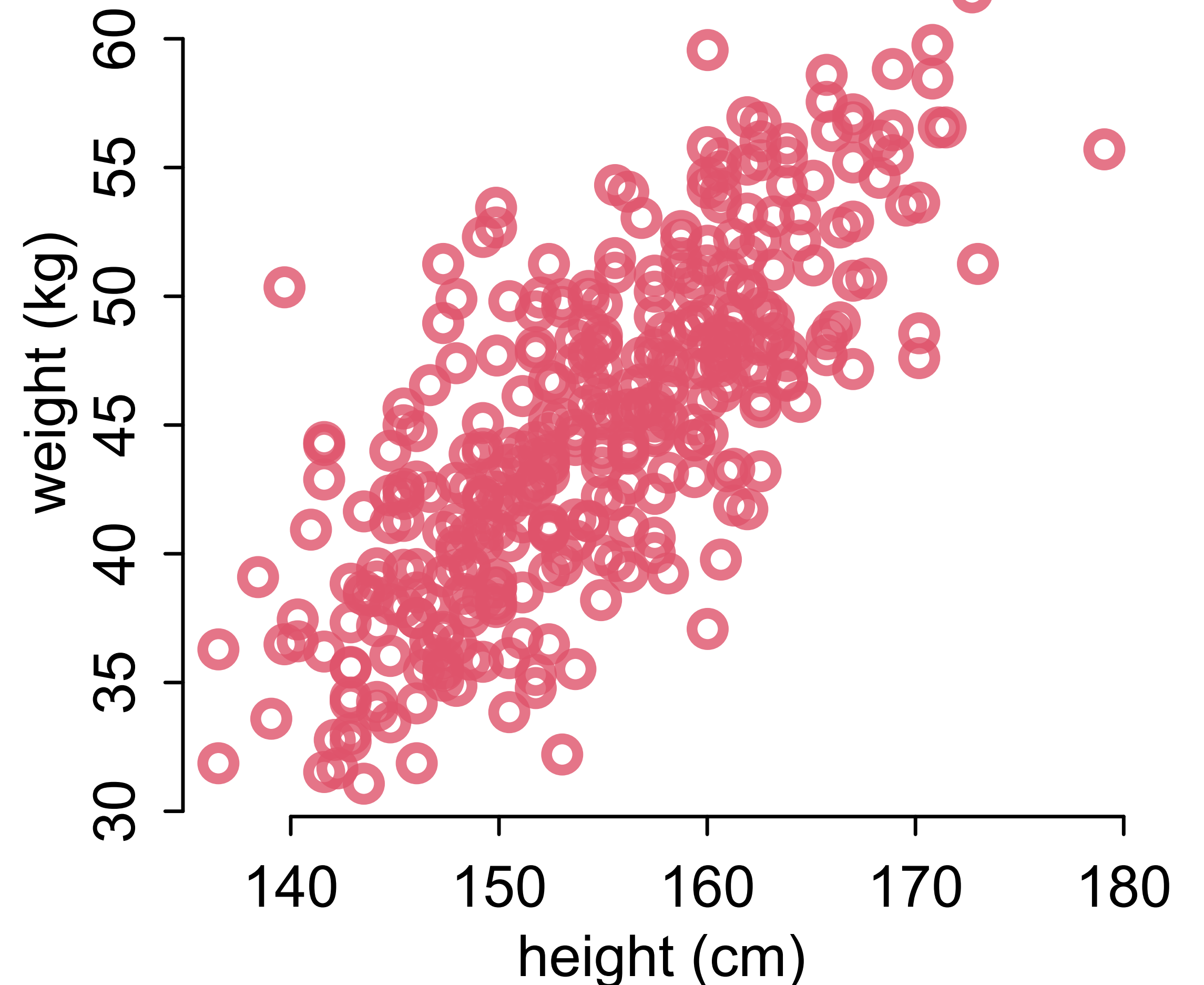
How does **height** influence **weight**?

$$H \longrightarrow W$$

$$W = f(H)$$

“Weight is some function of height”

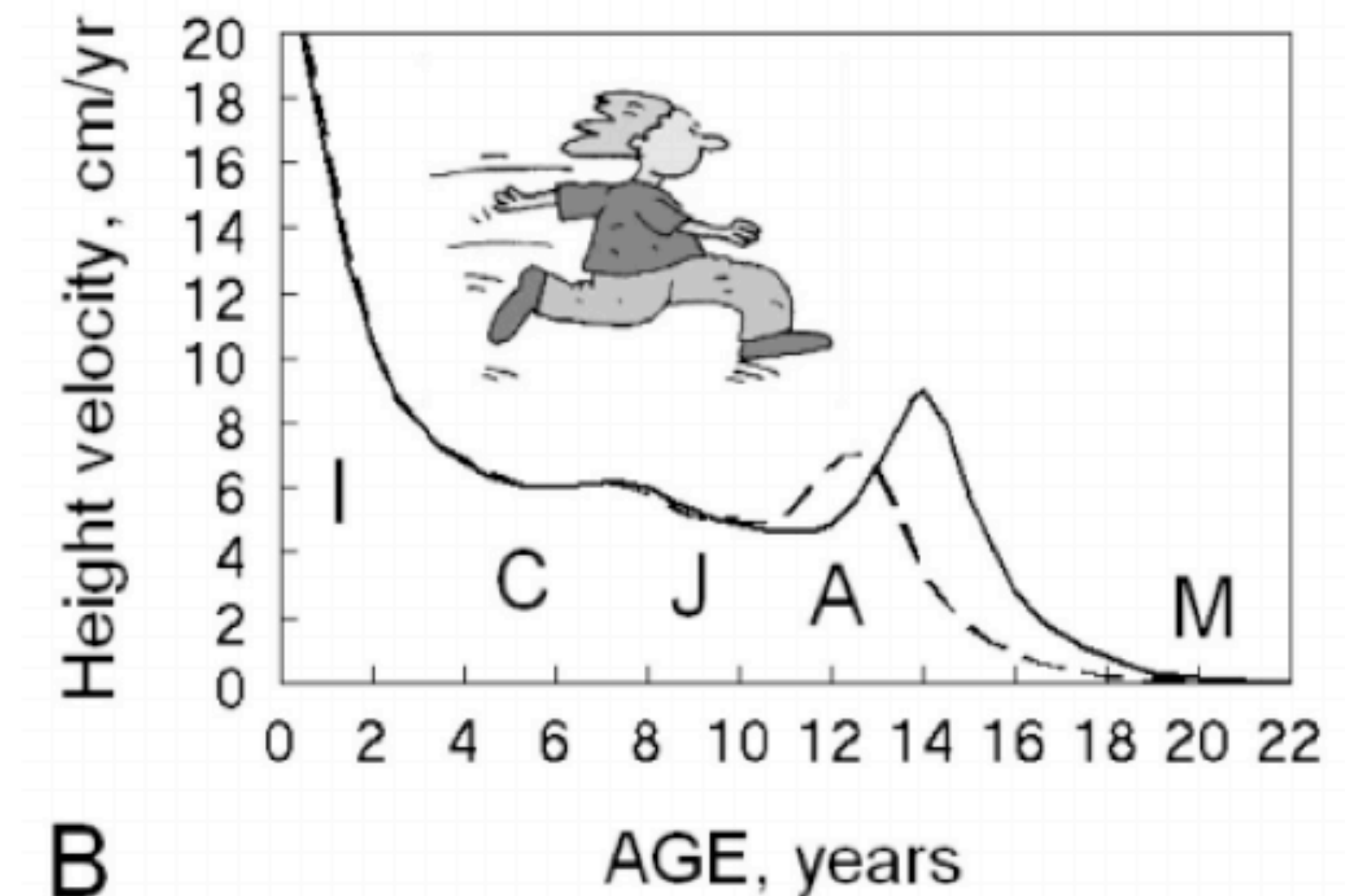
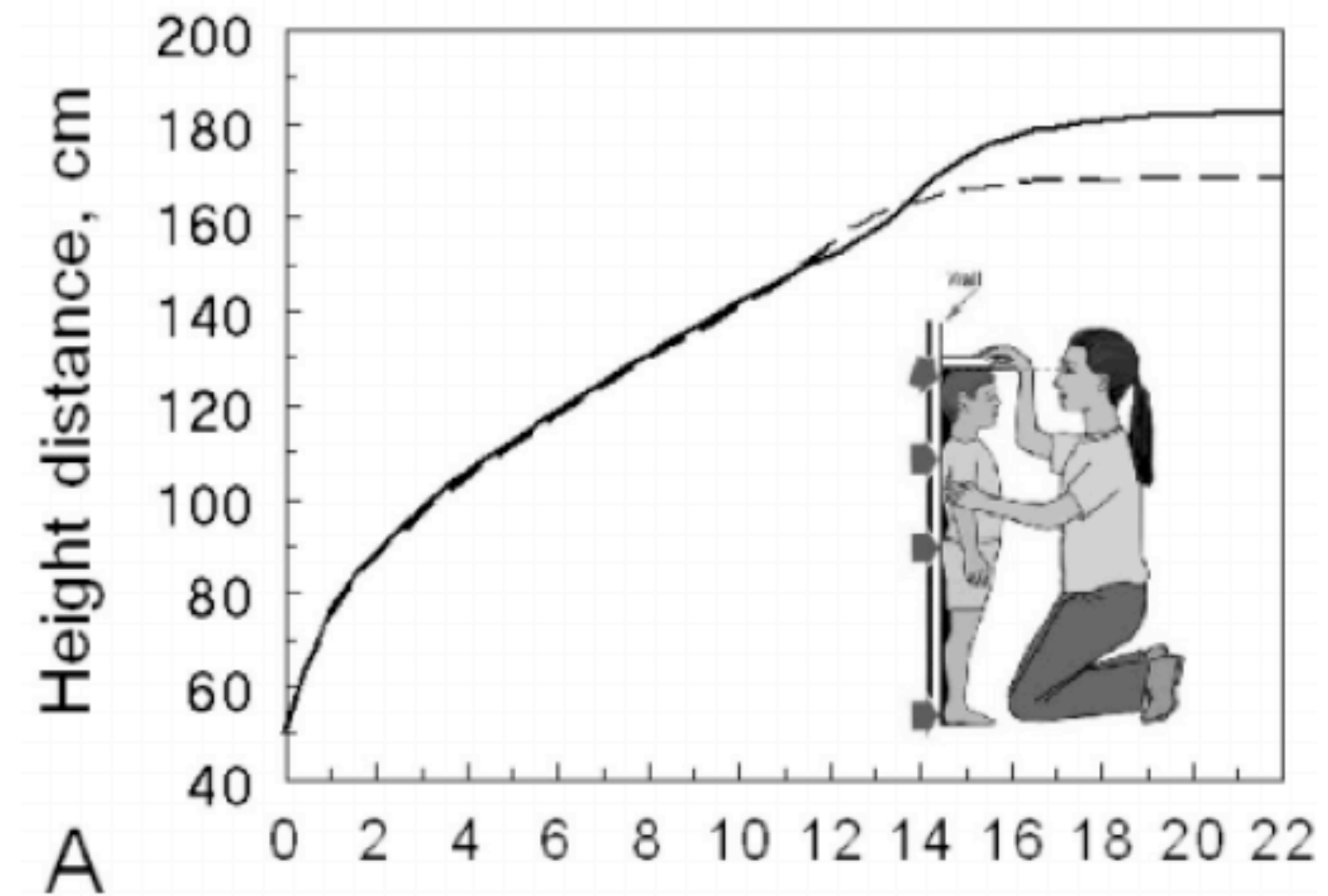
```
data(Howell1)  
d <- Howell1[Howell1$age >= 18,]
```



Generative models

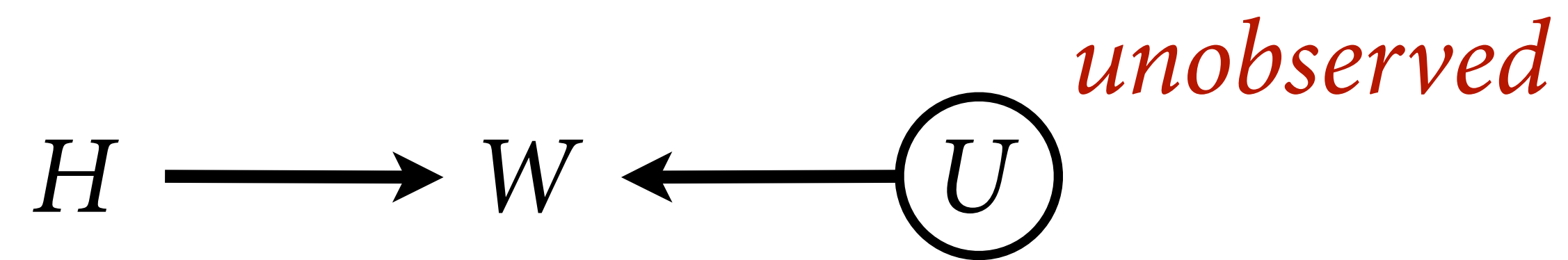
Options

- (1) **Dynamic**: Incremental growth of organism; both mass and height (length) derive from growth pattern; Gaussian variation result of summed fluctuations
- (2) **Static**: Changes in height result in changes in weight, but no mechanism; Gaussian variation result of growth history



(2) Scientific model

How does **height** influence **weight**?



$$W = f(H, U)$$

“Weight is some function of height and unobserved stuff”

Generative model: $H \rightarrow W$



For adults, weight is a proportion of height plus the influence of unobserved causes:

$$W = \beta H + U$$

Generative model: $H \rightarrow W$

$$W = \beta H + U$$

Generative code:

R code
3.2

```
# function to simulate weights of individuals from height
sim_weight <- function(H,b,sd) {
  U <- rnorm( length(H) , 0 , sd )
  W <- b*H + U
  return(W)
}
```

R code
3.2

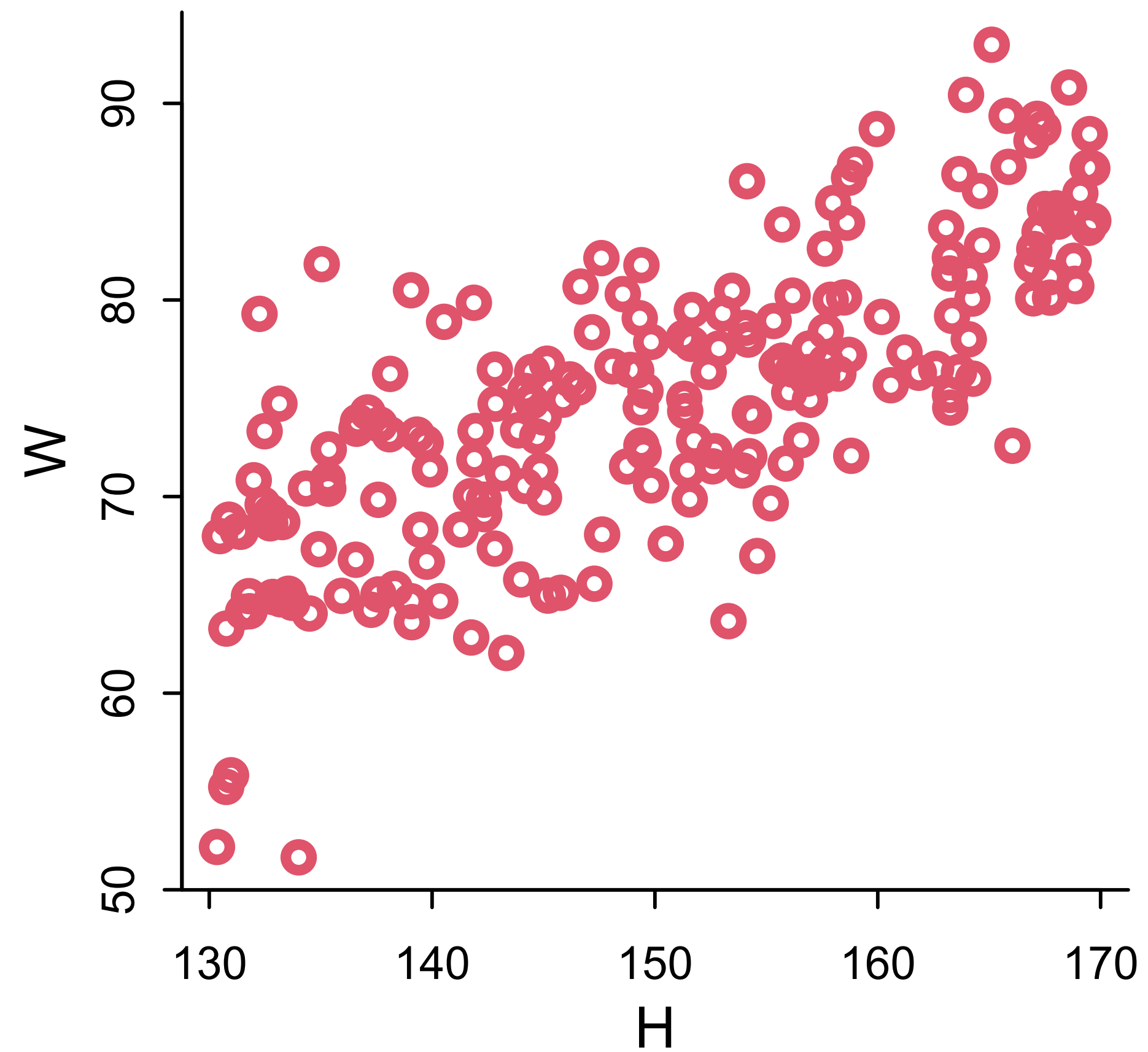
```
# function to simulate weights of individuals from height
sim_weight <- function(H,b,sd) {
  U <- rnorm( length(H) , 0 , sd )
  W <- b*H + U
  return(W)
}
```

R code
3.3

```
H <- runif( 200 , min=130 , max=170 )
W <- sim_weight( H , b=0.5 , sd=5 )
plot( W ~ H , col=2 , lwd=3 )
```


R code
3.3

```
H <- runif( 200 , min=130 , max=170 )  
W <- sim_weight( H , b=0.5 , sd=5 )  
plot( W ~ H , col=2 , lwd=3 )
```



Describing models

Conventional statistical model notation:

(1) List the variables

(2) Define each variable as a deterministic or distributional function of the other variables

Describing models

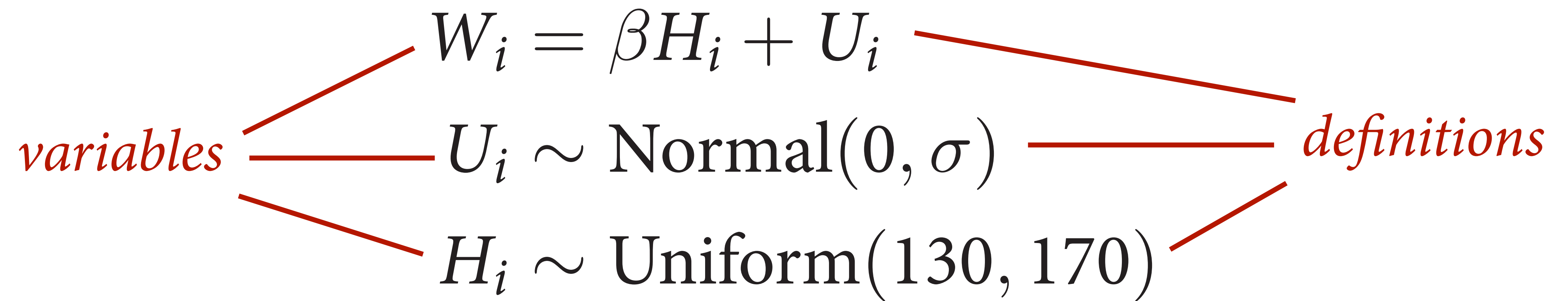
R code
3.2

```
# function to simulate weights of individuals from height
sim_weight <- function(H,b,sd) {
  U <- rnorm( length(H) , 0 , sd )
  W <- b*H + U
  return(W)
}
```

$$W_i = \beta H_i + U_i$$

$$U_i \sim \text{Normal}(0, \sigma)$$

$$H_i \sim \text{Uniform}(130, 170)$$



$$W_i = \beta H_i + U_i$$

$$U_i \sim \text{Normal}(0, \sigma)$$

$$H_i \sim \text{Uniform}(130, 170)$$

individuals

deterministic

$$W_i = \beta H_i + U_i$$

$$U_i \sim \text{Normal}(0, \sigma)$$

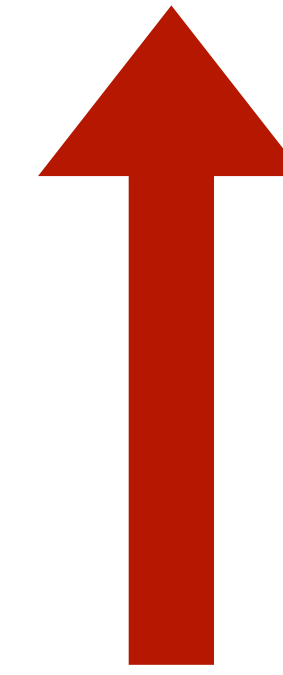
$$H_i \sim \text{Uniform}(130, 170)$$

distributed as

$$W_i = \beta H_i + U_i$$

$$U_i \sim \text{Normal}(0, \sigma)$$

$$H_i \sim \text{Uniform}(130, 170)$$



R code
3.2

```
# function to simulate weights of individuals from height
sim_weight <- function(H,b,sd) {
  U <- rnorm( length(H) , 0 , sd )
  W <- b*H + U
  return(W)
}
```



PAUSE

Linear Regression

```
data(Howell1)  
d <- Howell1[Howell1$age >= 18,]
```

Drawing the Owl

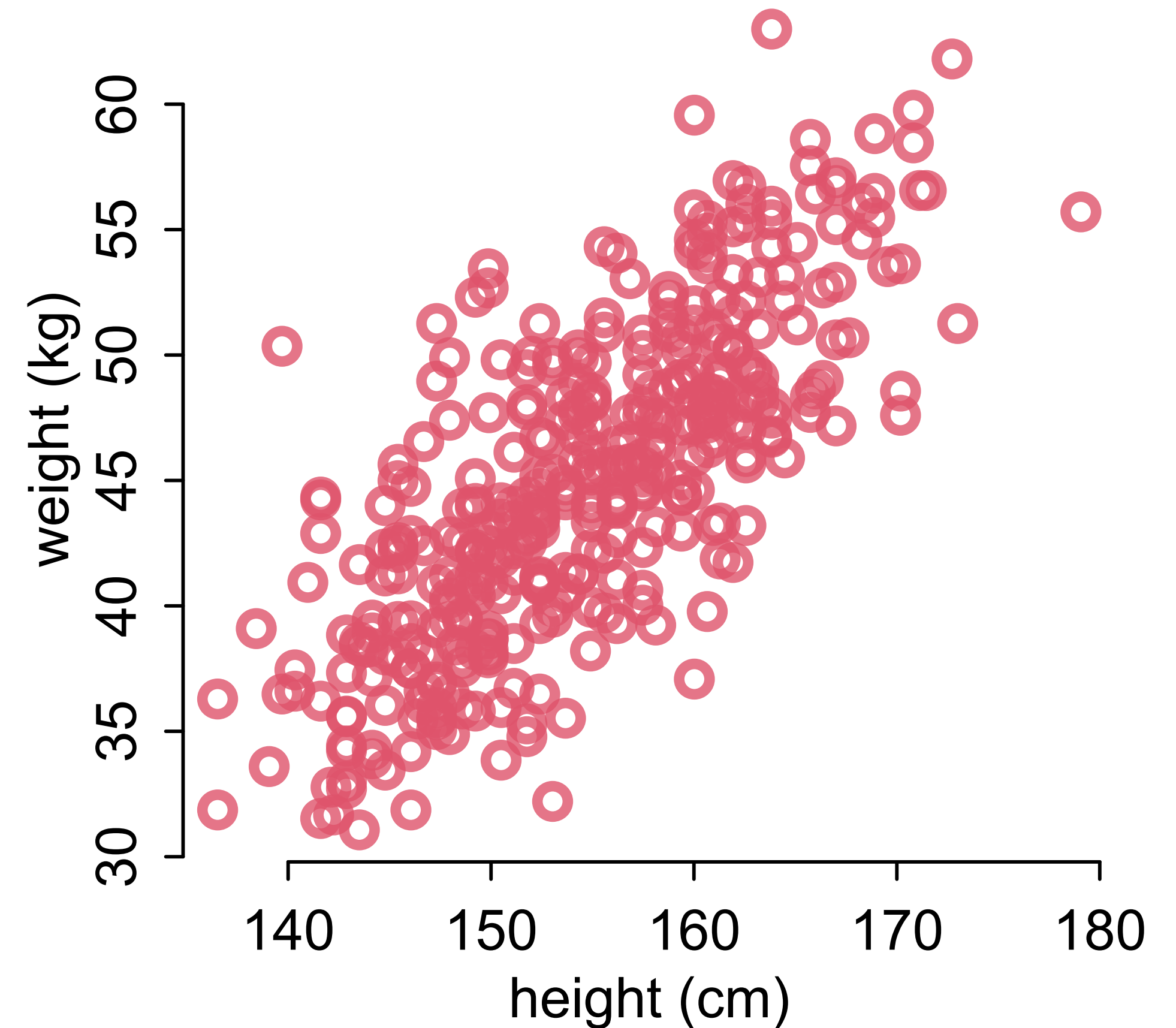
(1) Question/goal/estimand

(2) Scientific model

(3) Statistical model(s)

(4) Validate model

(5) Analyze data



Estimator

We want to estimate how the average weight changes with height.

$$E(W_i | H_i) = \alpha + \beta H_i$$

*Average weight
conditional on height*

intercept

slope

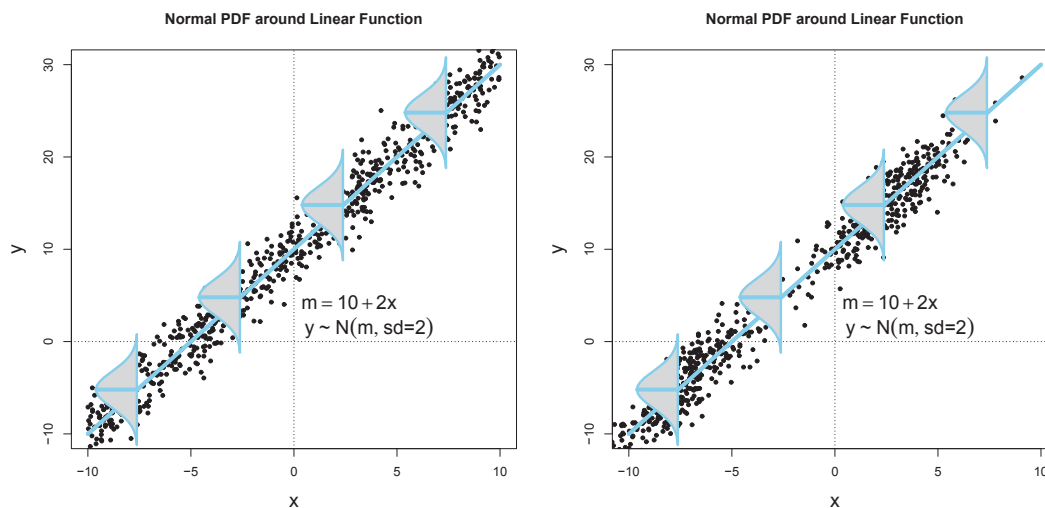


Figure 17.1: Examples of points normally distributed around a linear function. (The left panel repeats Figure 15.9, p. 406.) The model assumes that the data y are normally distributed vertically around the line, as shown. Moreover, the variance of y is the same at all values of x . The model puts no constraints on the distribution of x . The right panel shows a case in which x are distributed bimodally, whereas in the left panel the x are distributed uniformly. In both panels, there is homogeneity of variance. Copyright © Kruschke, J. K. (2014). *Doing Bayesian Data Analysis: A Tutorial with R, JAGS, and Stan. 2nd Edition*. Academic Press / Elsevier.

Posterior distribution

$$\Pr(\alpha, \beta, \sigma | H_i, W_i) = \frac{\Pr(W_i | H_i, \alpha, \beta, \sigma) \Pr(\alpha, \beta, \sigma)}{Z}$$

Posterior distribution

$$\Pr(\alpha, \beta, \sigma | H_i, W_i) = \frac{\Pr(W_i | H_i, \alpha, \beta, \sigma) \Pr(\alpha, \beta, \sigma)}{Z}$$

*posterior probability
of specific line*

Posterior distribution

garden of forking data

$$\Pr(\alpha, \beta, \sigma | H_i, W_i) = \frac{\Pr(W_i | H_i, \alpha, \beta, \sigma) \Pr(\alpha, \beta, \sigma)}{Z}$$

*posterior probability
of specific line*

Posterior distribution

$$\Pr(\alpha, \beta, \sigma | H_i, W_i) = \frac{\Pr(W_i | H_i, \alpha, \beta, \sigma) \Pr(\alpha, \beta, \sigma)}{Z}$$

garden of forking data

prior

*posterior probability
of specific line*

Posterior distribution

$$\Pr(\alpha, \beta, \sigma | H_i, W_i) = \frac{\Pr(W_i | H_i, \alpha, \beta, \sigma) \Pr(\alpha, \beta, \sigma)}{Z}$$

garden of forking data

prior

posterior probability of specific line

normalizing constant

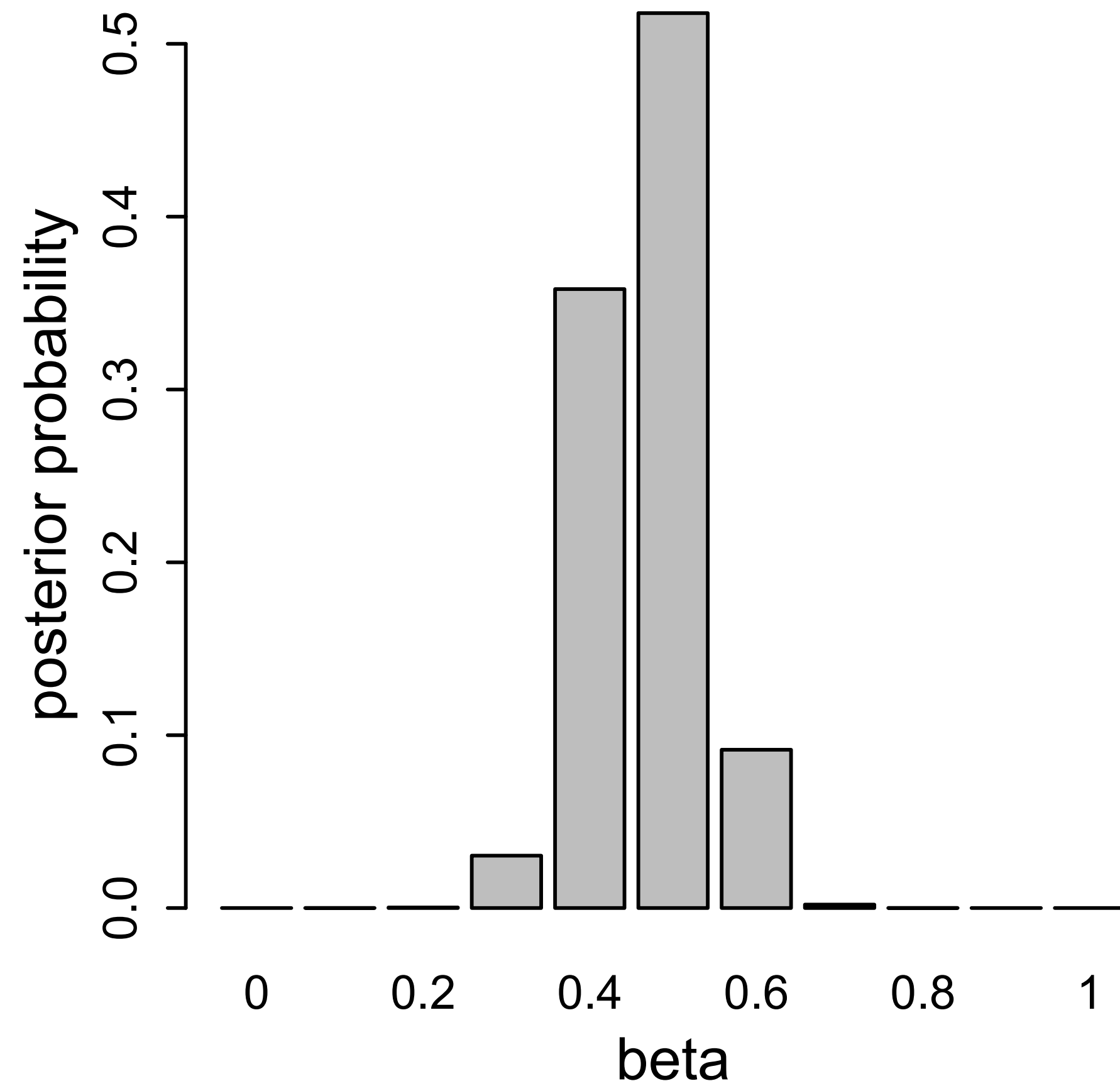
$$\Pr(\alpha, \beta, \sigma | H_i, W_i) = \frac{\Pr(W_i | H_i, \alpha, \beta, \sigma) \Pr(\alpha, \beta, \sigma)}{Z}$$

$$W_i \sim \text{Normal}(\mu_i, \sigma)$$

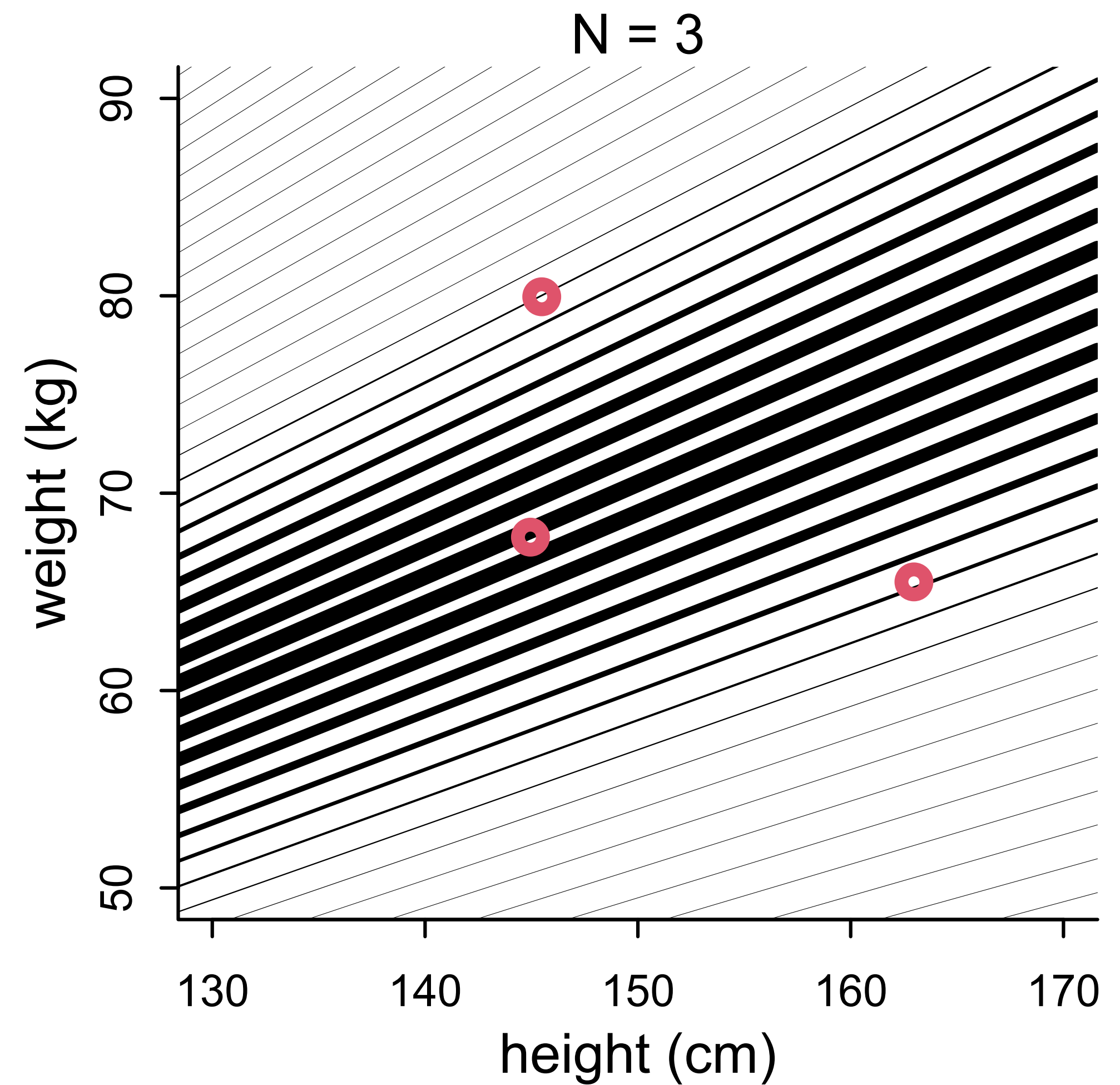
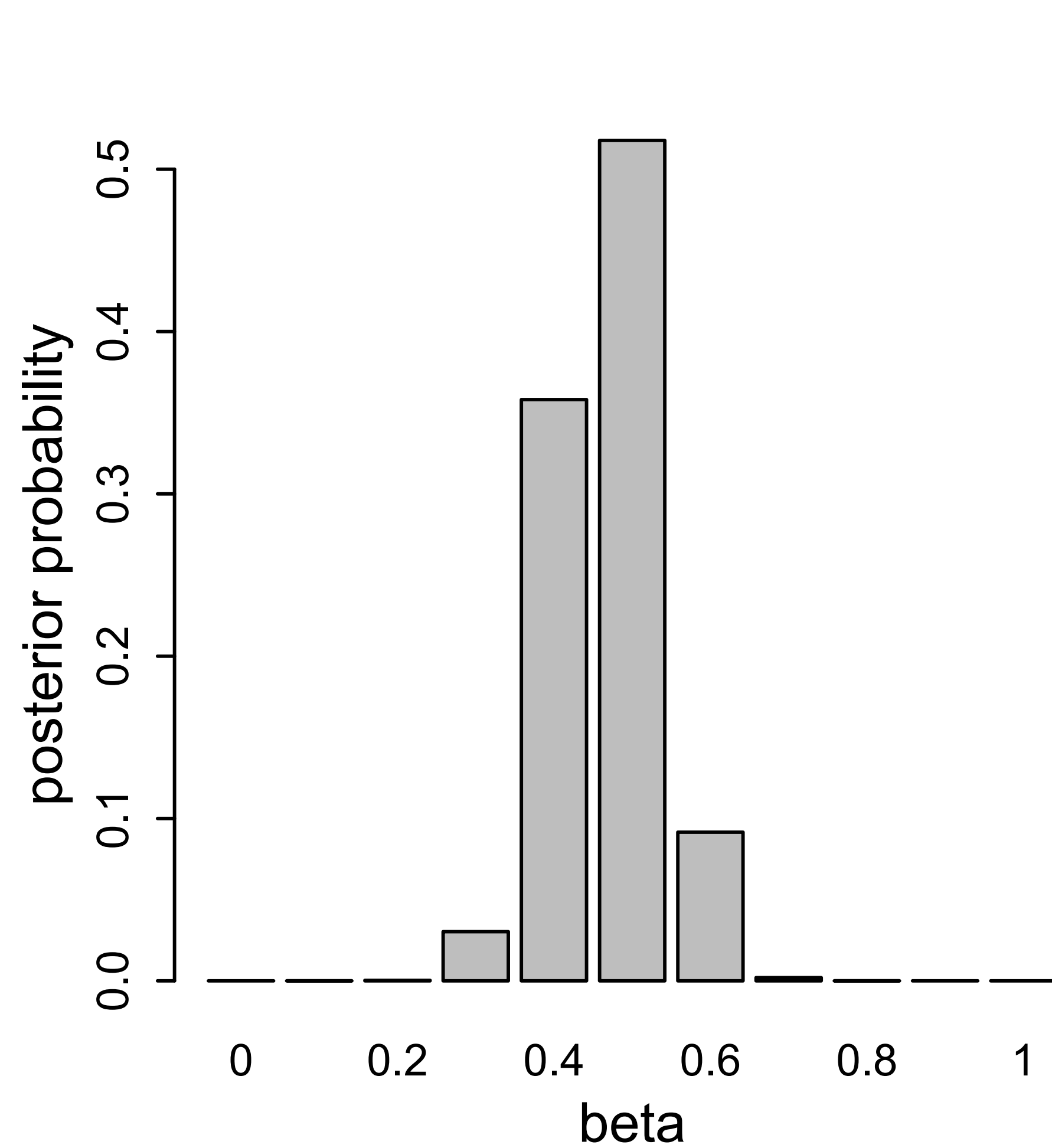
$$\mu_i = \alpha + \beta H_i$$

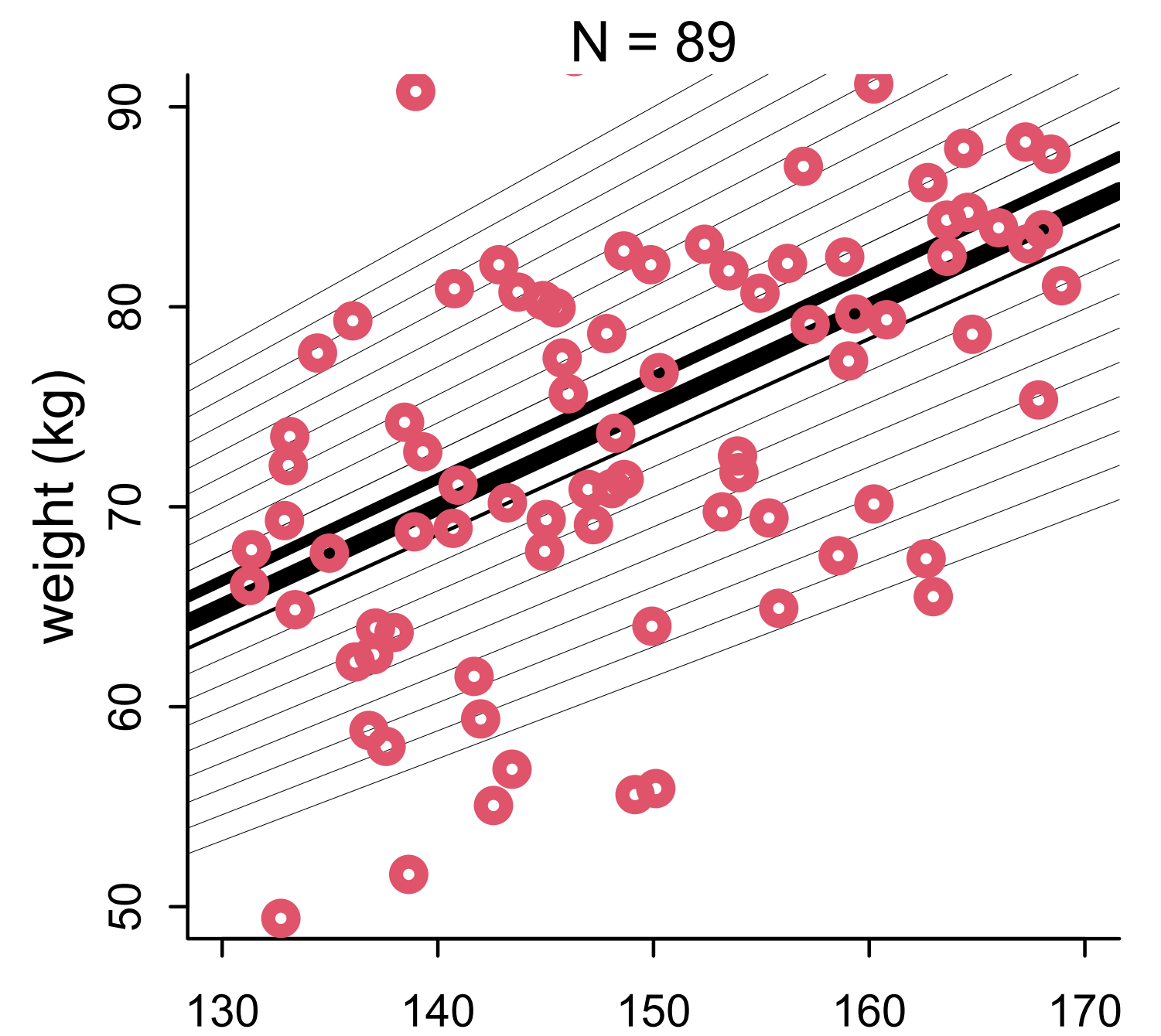
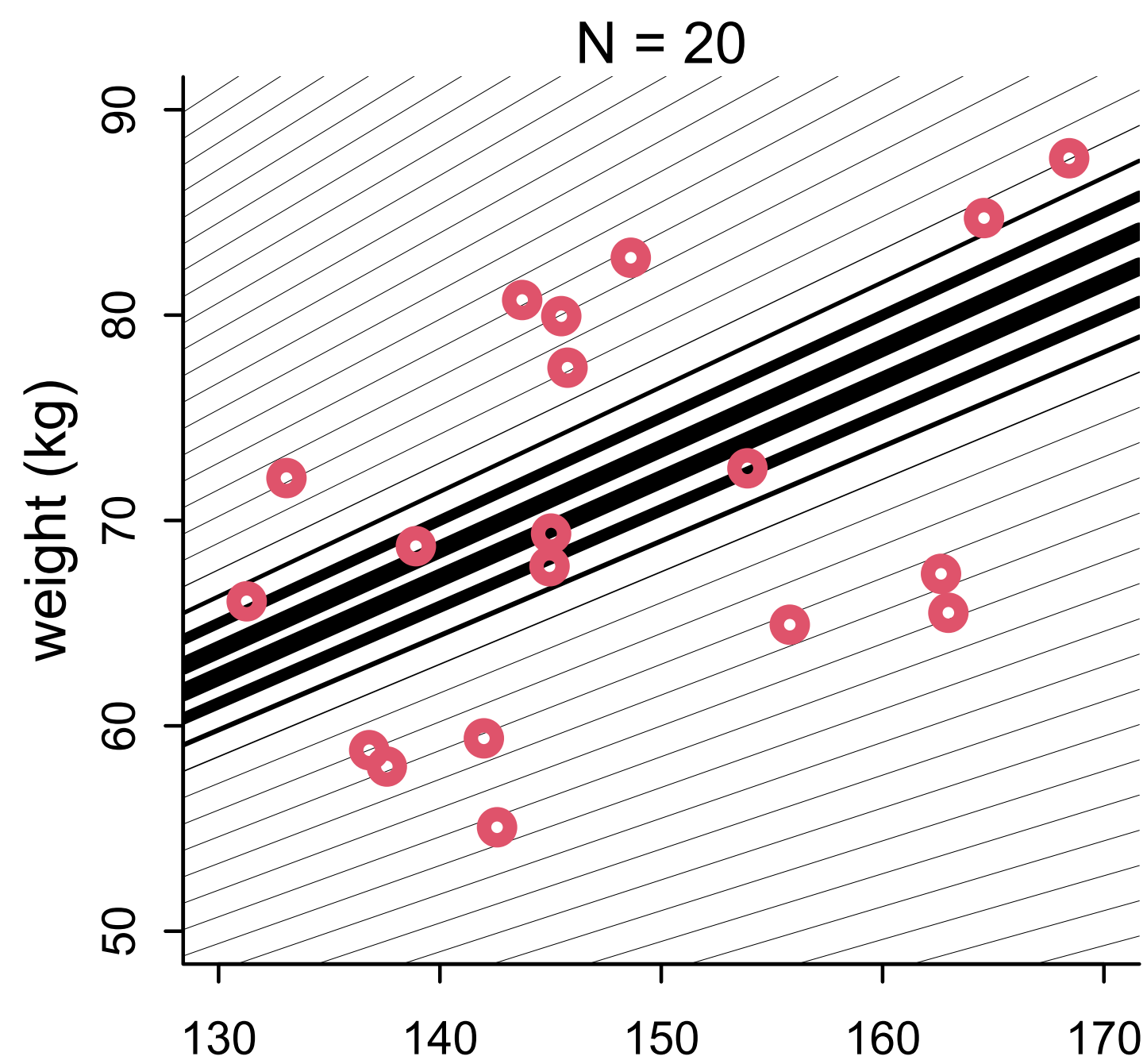
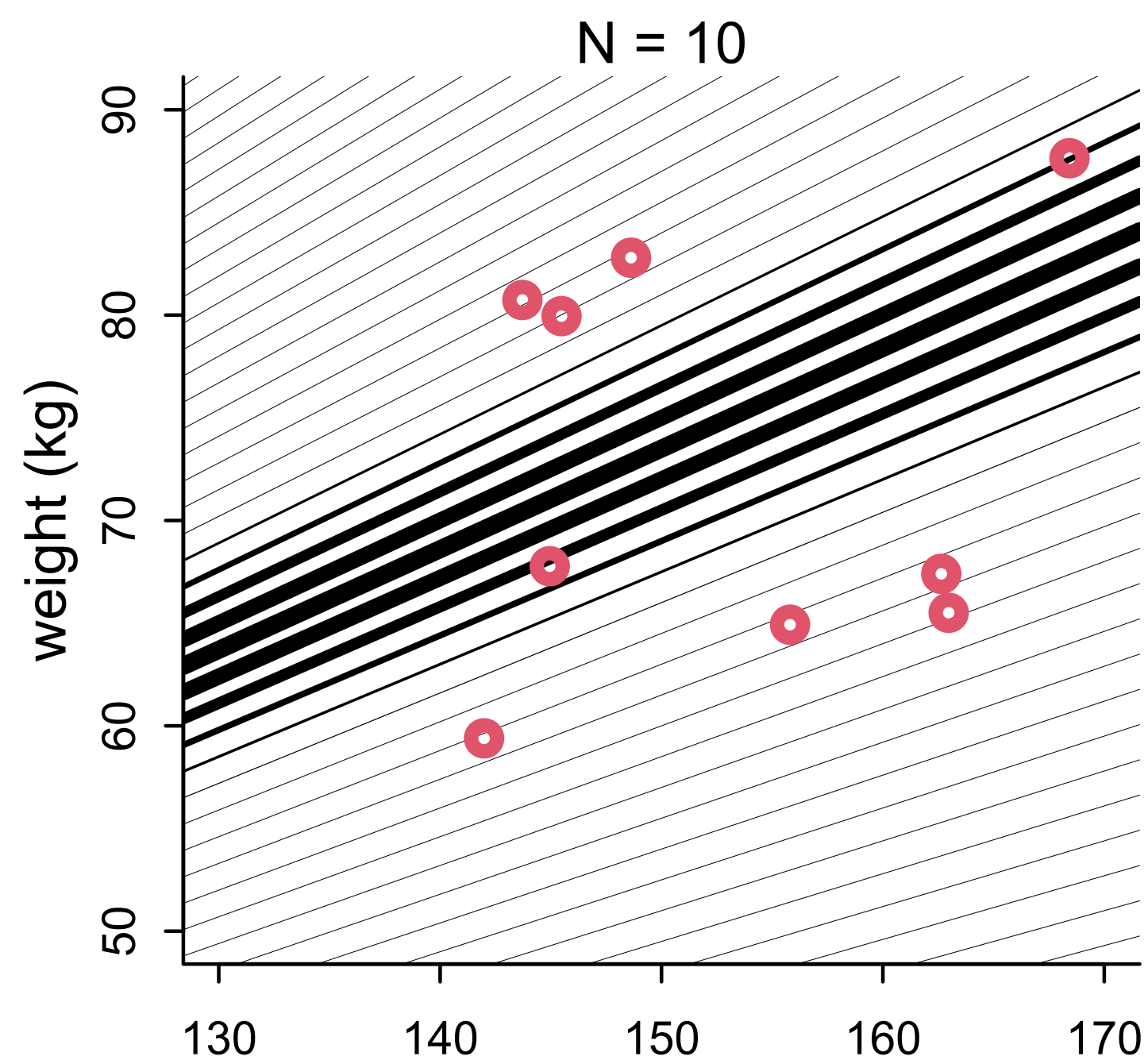
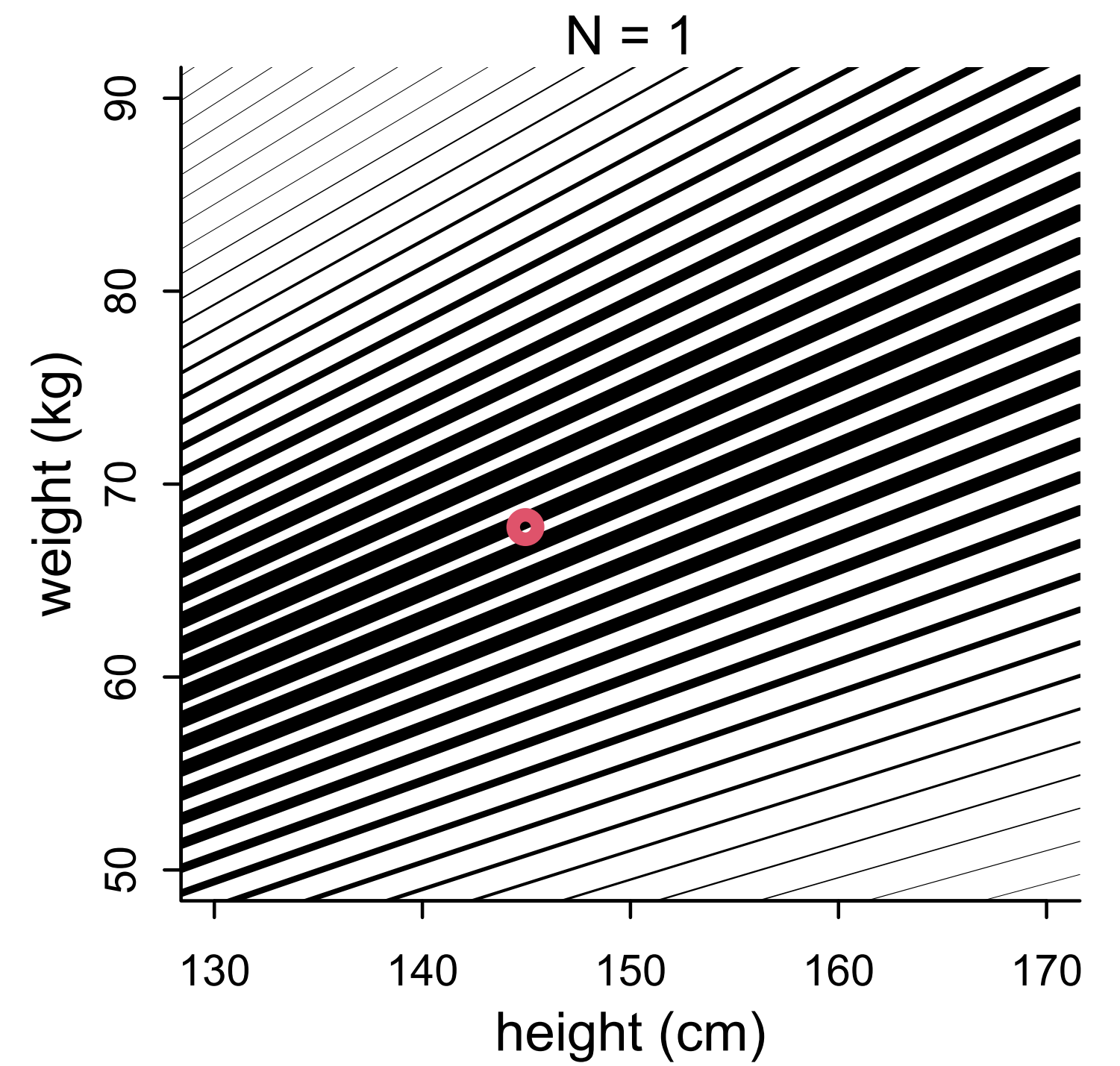
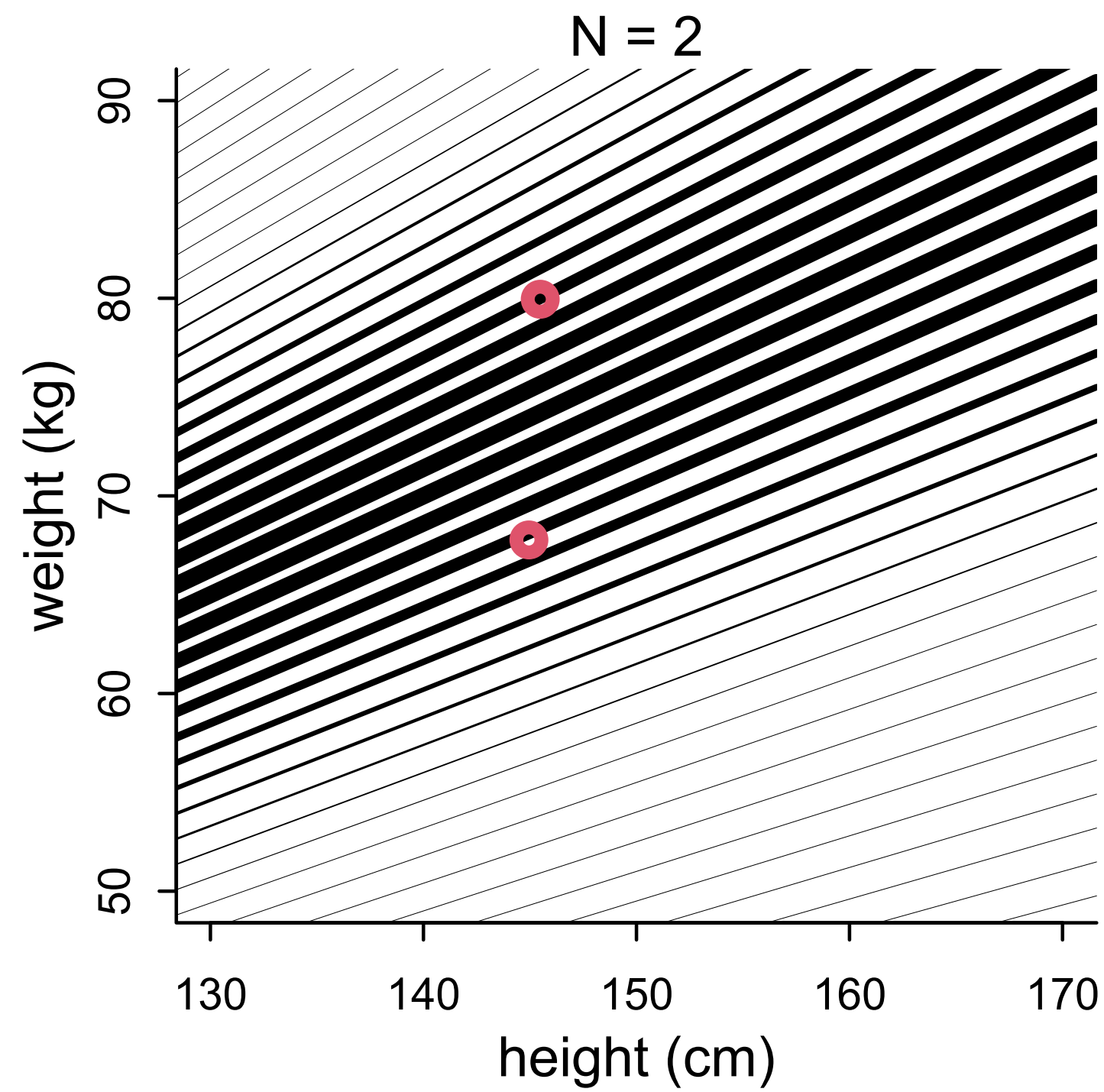
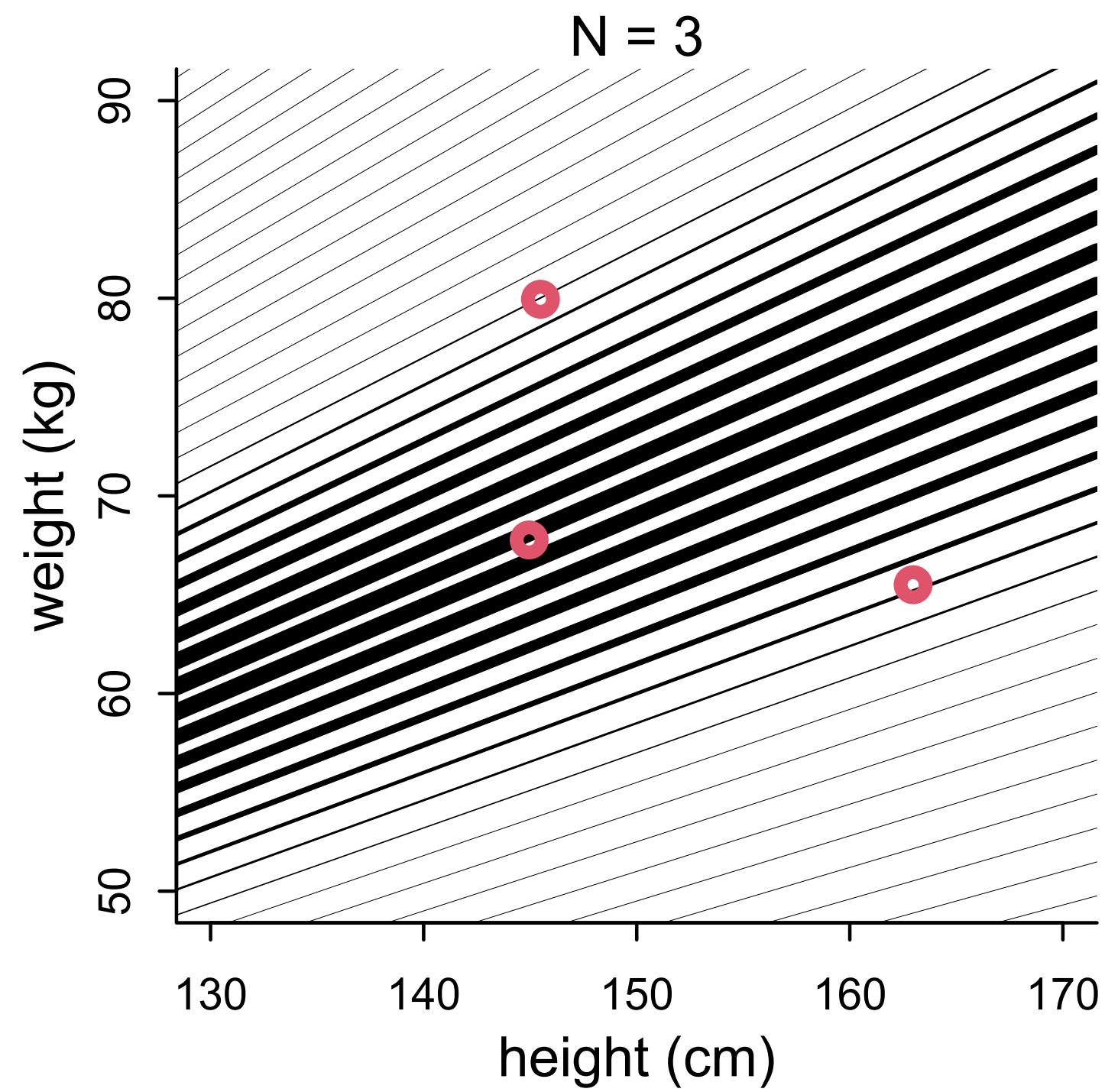
W is distributed normally with mean that is a linear function of H

Grid approximate posterior

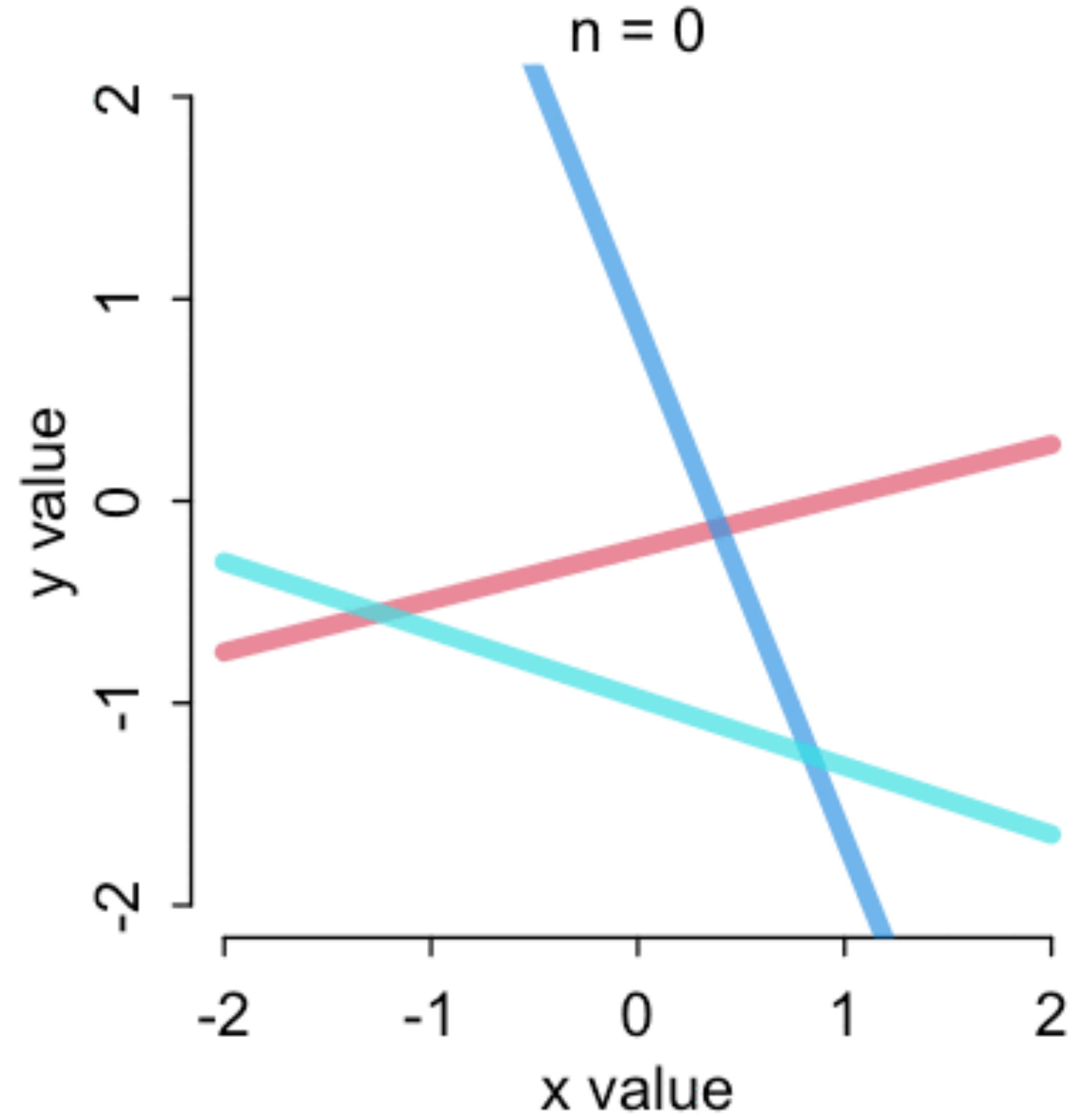
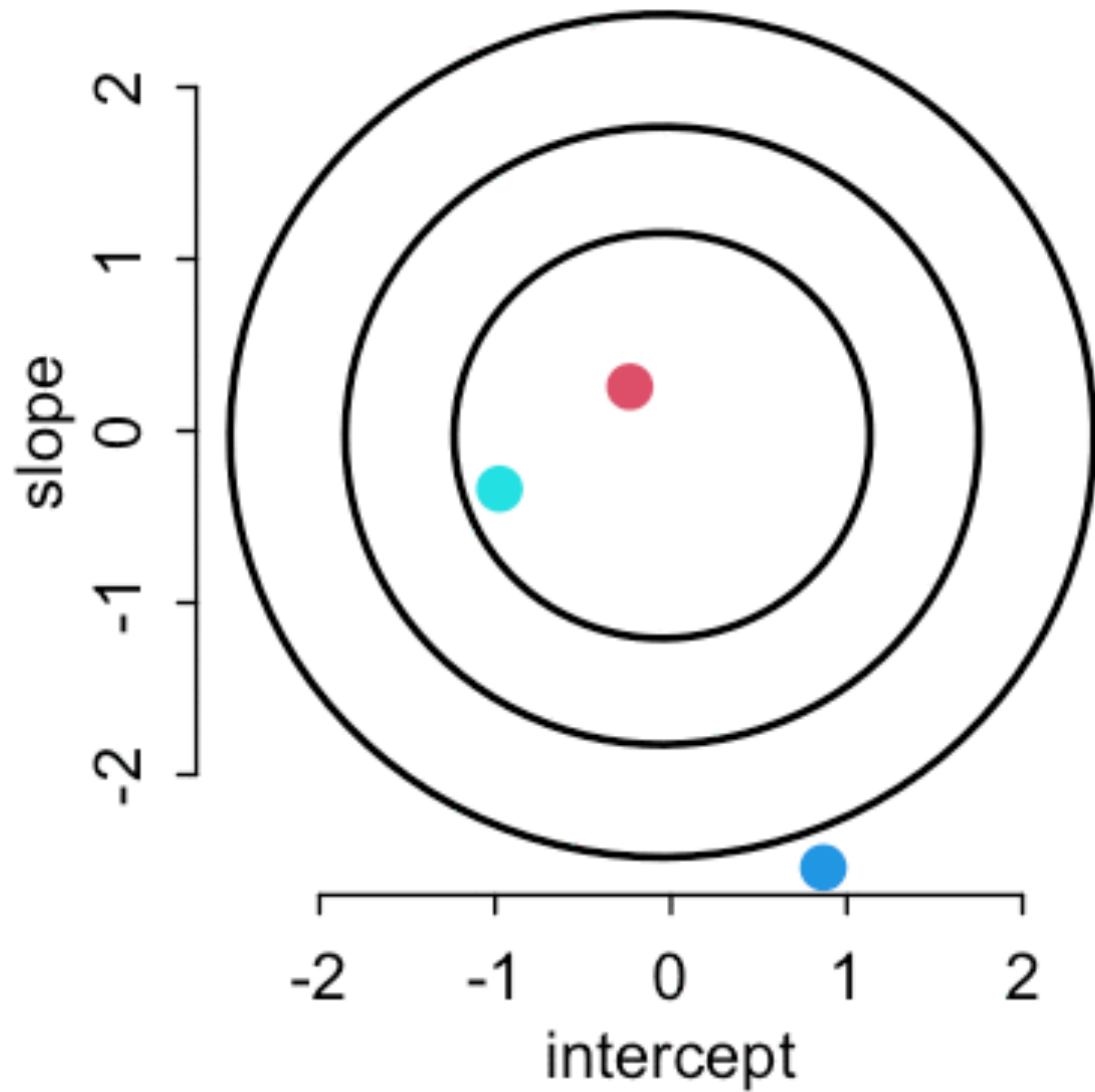


Grid approximate posterior

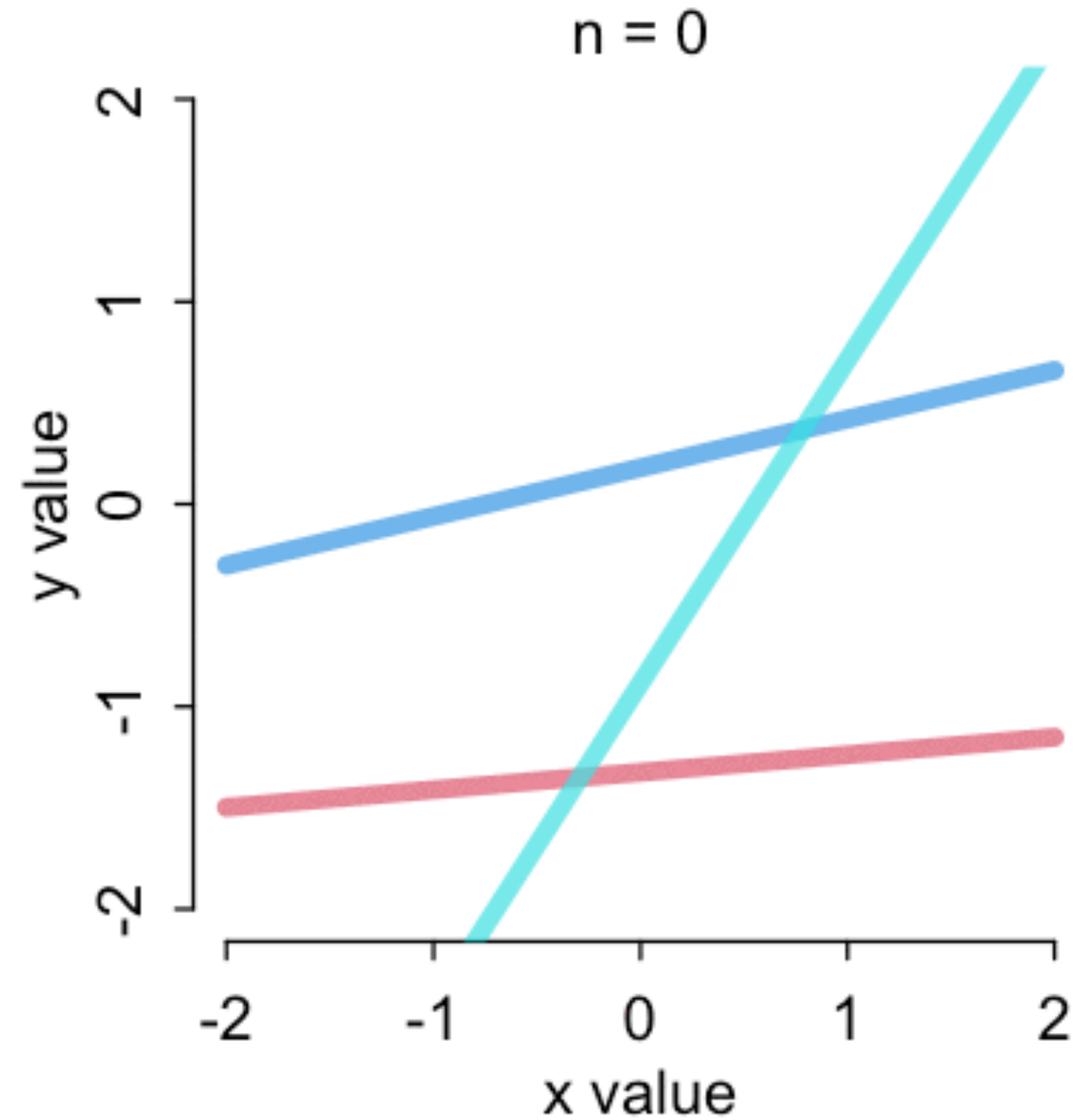
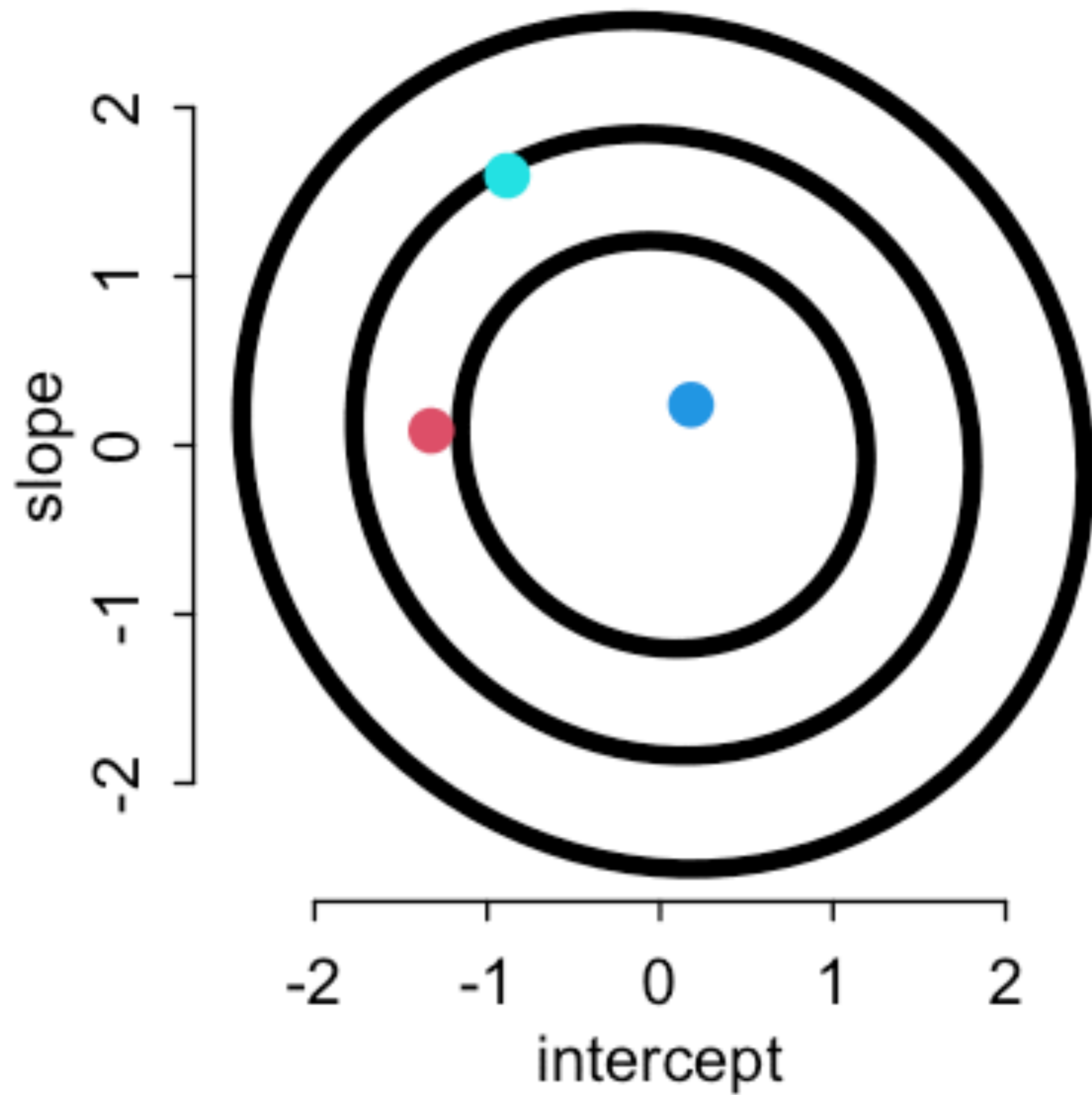




Updating the posterior



Updating the posterior



Enough grid approximation

We'll use quadratic approximation for the rest of the first half of the course.

$$W_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta H_i$$

$$\alpha \sim \text{Normal}(0, 10)$$

$$\beta \sim \text{Uniform}(0, 1)$$

$$\sigma \sim \text{Uniform}(0, 10)$$

```
m3.1 <- quap(  
  alist(  
    W ~ dnorm(mu,sigma),  
    mu <- a + b*H,  
    a ~ dnorm(0,10),  
    b ~ dunif(0,1),  
    sigma ~ dunif(0,10)  
  ) , data=list(W=W,H=H) )
```

Enough grid approximation

We'll use quadratic approximation for the rest of the first half of the course.

$$W_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta H_i$$

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```
m3.1 <- quap(  
  alist(  
    W ~ dnorm(mu,sigma),  
    mu <- a + b*H,  
    a ~ dnorm(0,10),  
    b ~ dunif(0,1),  
    sigma ~ dunif(0,10)  
  ) , data=list(W=W,H=H) )
```


Prior predictive distribution

Priors should express scientific knowledge, but **softly**

When $H = 0$, $W = 0$

Weight increases (on avg) with height

Weight (kg) is less than height (cm)

sigma must be positive

$$W_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta H_i$$

$$\alpha \sim \text{Normal}(0, 10)$$

$$\beta \sim \text{Uniform}(0, 1)$$

$$\sigma \sim \text{Uniform}(0, 10)$$

Prior predictive distribution

Understand the implications of priors through simulation

What do the observable variables look like with these priors?

$$W_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta H_i$$

$$\alpha \sim \text{Normal}(0, 10)$$

$$\beta \sim \text{Uniform}(0, 1)$$

$$\sigma \sim \text{Uniform}(0, 10)$$

```
n <- 1e3
a <- rnorm(n,0,10)
b <- runif(n,0,1)
plot( NULL , xlim=c(130,170) , ylim=c(50,90) ,
      xlab="height (cm)" , ylab="weight (kg)" )
for ( j in 1:50 ) abline( a=a[j] , b=b[j] , lwd=2 , col=2 )
```

$$W_i \sim \text{Normal}(\mu_i, \sigma)$$

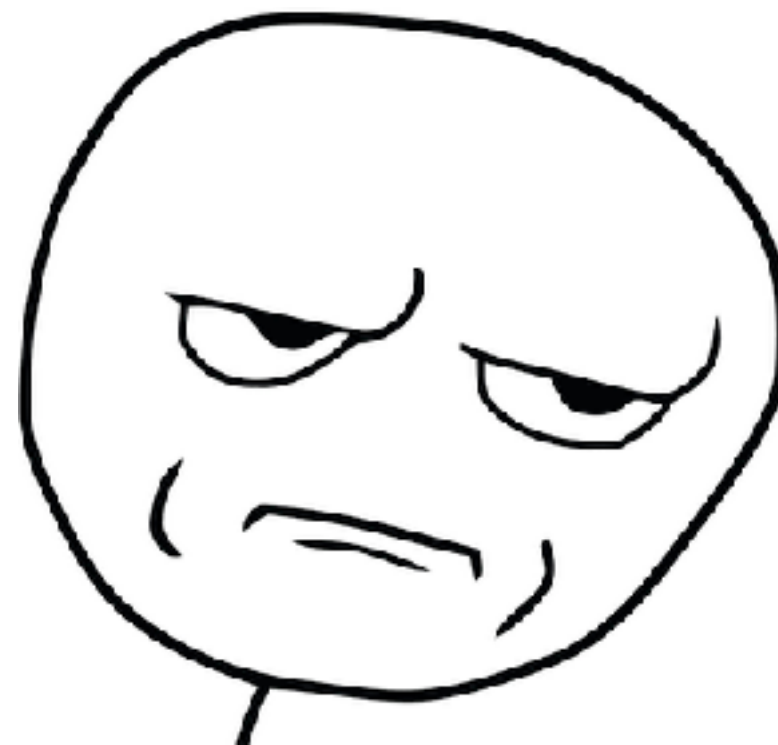
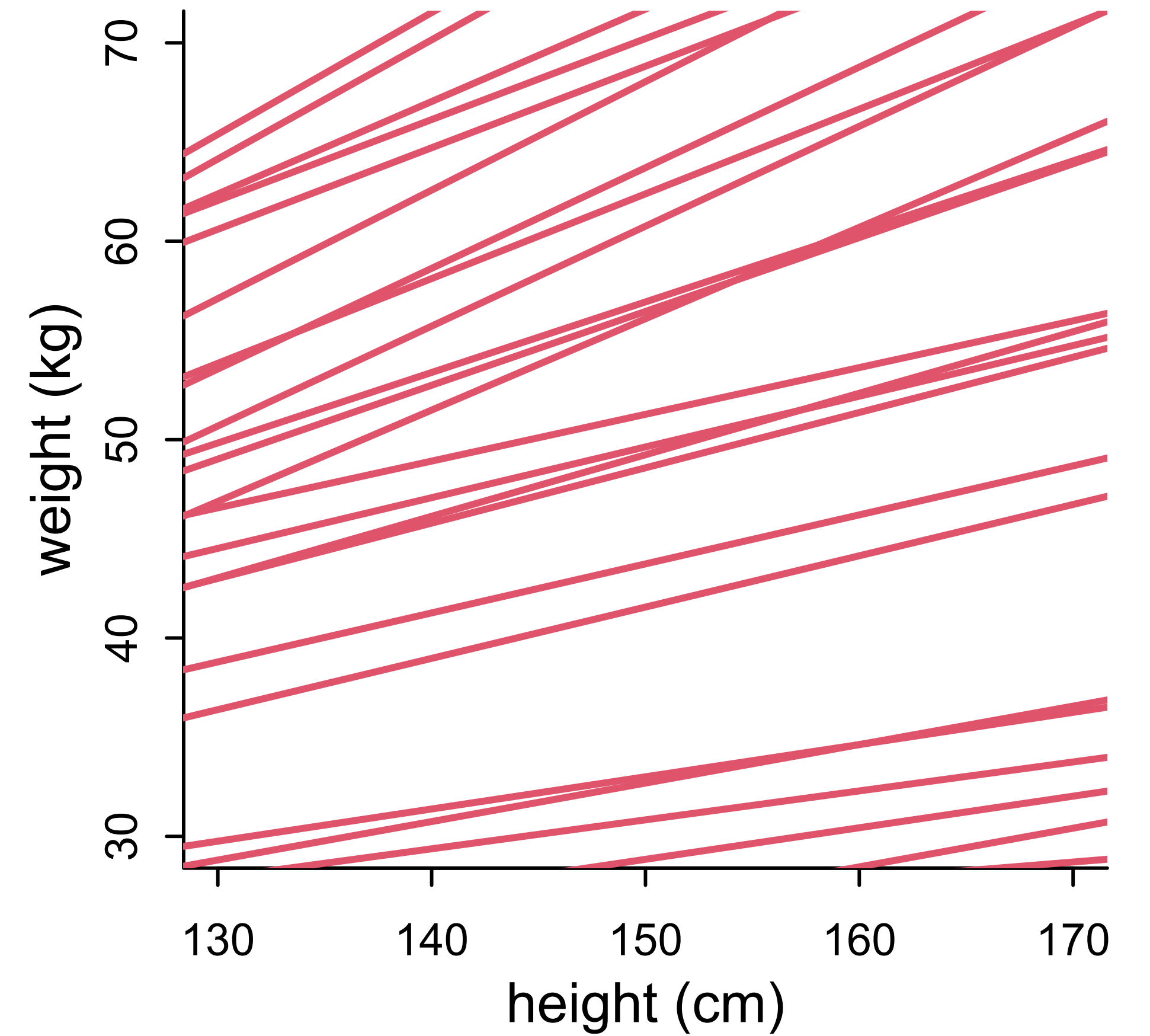
$$\mu_i = \alpha + \beta H_i$$

$$\alpha \sim \text{Normal}(0, 10)$$

$$\beta \sim \text{Uniform}(0, 1)$$

$$\sigma \sim \text{Uniform}(0, 10)$$


```
n <- 1e3
a <- rnorm(n,0,10)
b <- runif(n,0,1)
plot( NULL , xlim=c(130,170) , ylim=c(50,90) ,
      xlab="height (cm)" , ylab="weight (kg)" )
for ( j in 1:50 ) abline( a=a[j] , b=b[j] , lwd=2 , col=2 )
```



Sermon on priors

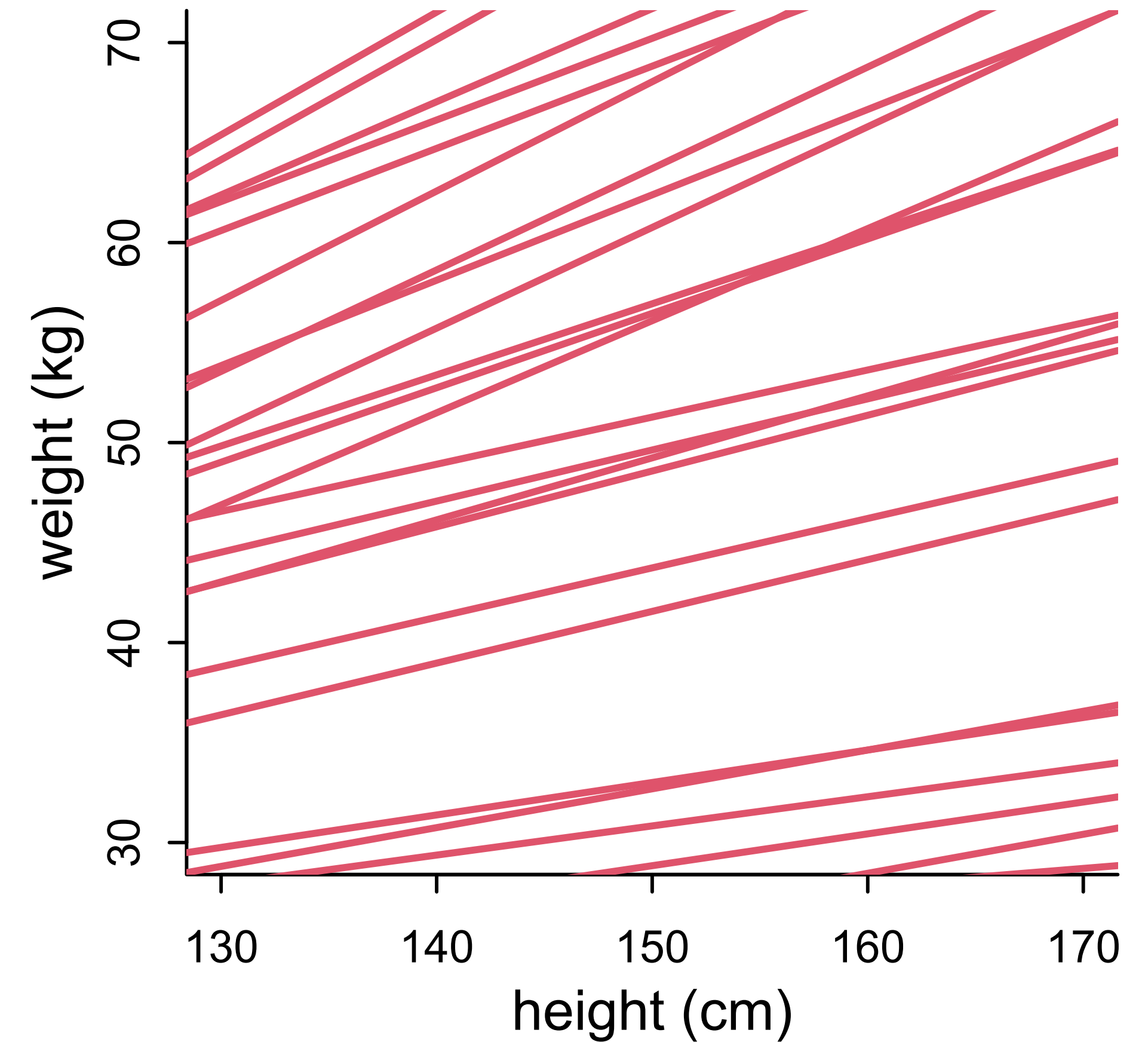
There are no correct priors, only scientifically justifiable priors

Justify with information outside the data —
like rest of model

Priors not so important in simple models

Very important/useful in complex models

Need to practice now: simulate, understand



Linear Regression

```
data(Howell1)  
d <- Howell1[Howell1$age>=18,]
```

Drawing the Owl

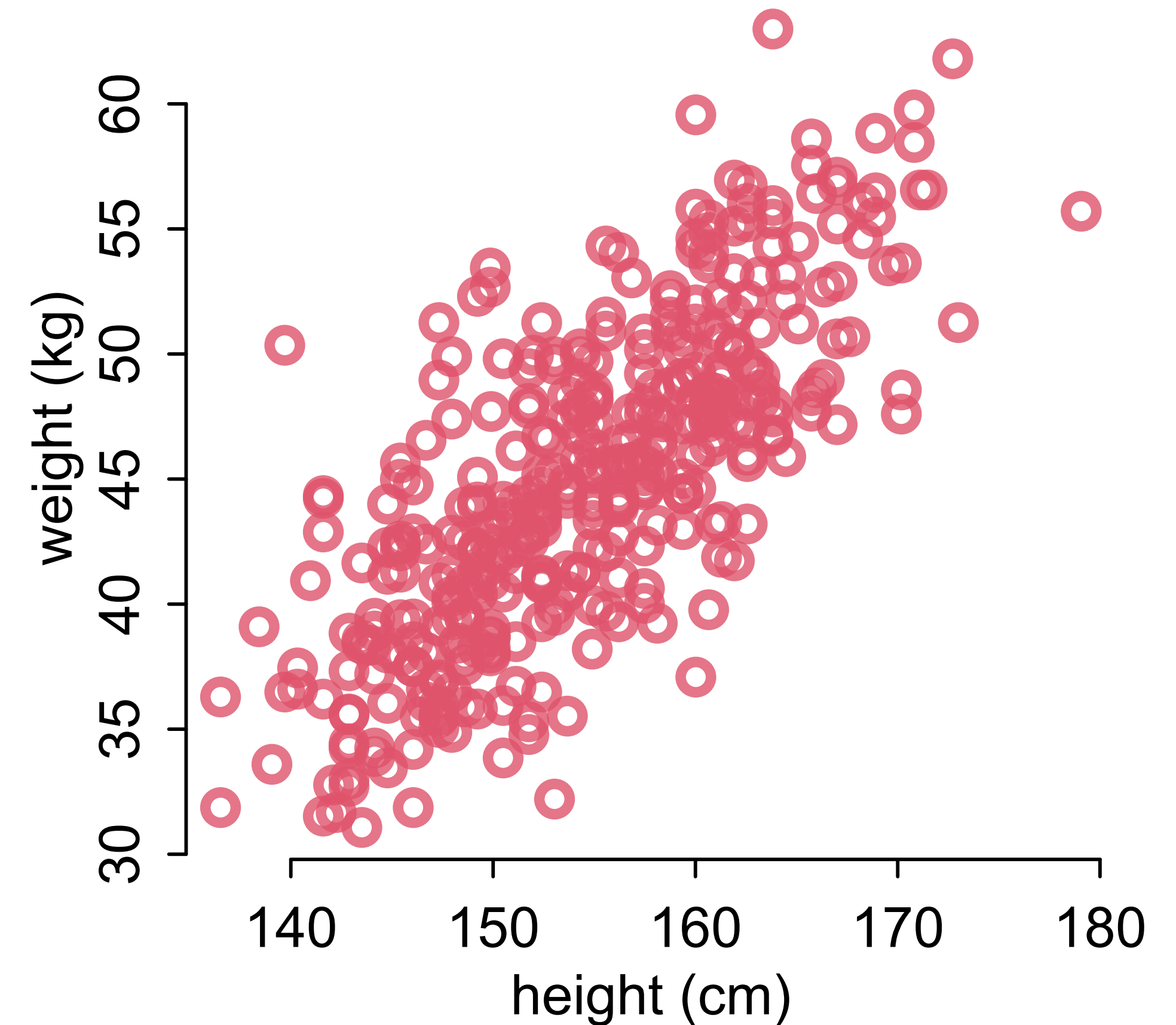
(1) Question/goal/estimand

(2) Scientific model

(3) Statistical model(s)

(4) Validate model

(5) Analyze data



Simulation-Based Validation

Bare minimum: Test statistical model with simulated observations from scientific model

Golem might be broken

Even working golems might not deliver what you hoped

Strong test: **Simulation-Based Calibration**



Fahrvergnügen


```
# simulate a sample of 10 people
set.seed(93)
H <- runif(10,130,170)
W <- sim_weight(H,b=0.5,sd=5)
```

```
# run the model
library(rethinking)
m3.1 <- quap(
  alist(
    W ~ dnorm(mu,sigma),
    mu <- a + b*H,
    a ~ dnorm(0,10),
    b ~ dunif(0,1),
    sigma ~ dunif(0,10)
  ) , data=list(W=W,H=H) )
```

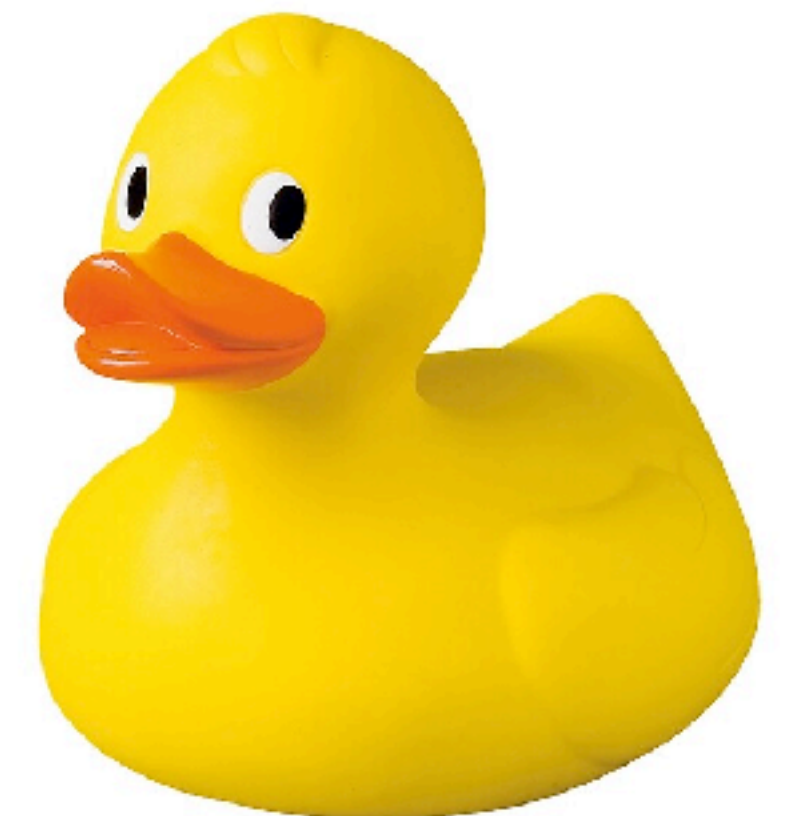
```
# summary
precis( m3.1 )
```

	mean	sd	5.5%	94.5%
a	5.19	9.43	-9.88	20.26
b	0.49	0.07	0.38	0.59
sigma	5.64	1.29	3.57	7.71

Vary slope and make sure
posterior mean tracks it

Use a large sample to see
that it converges to data
generating value

Same for other unknowns
(parameters)



Linear Regression

```
data(Howell1)  
d <- Howell1[Howell1$age>=18,]
```

Drawing the Owl

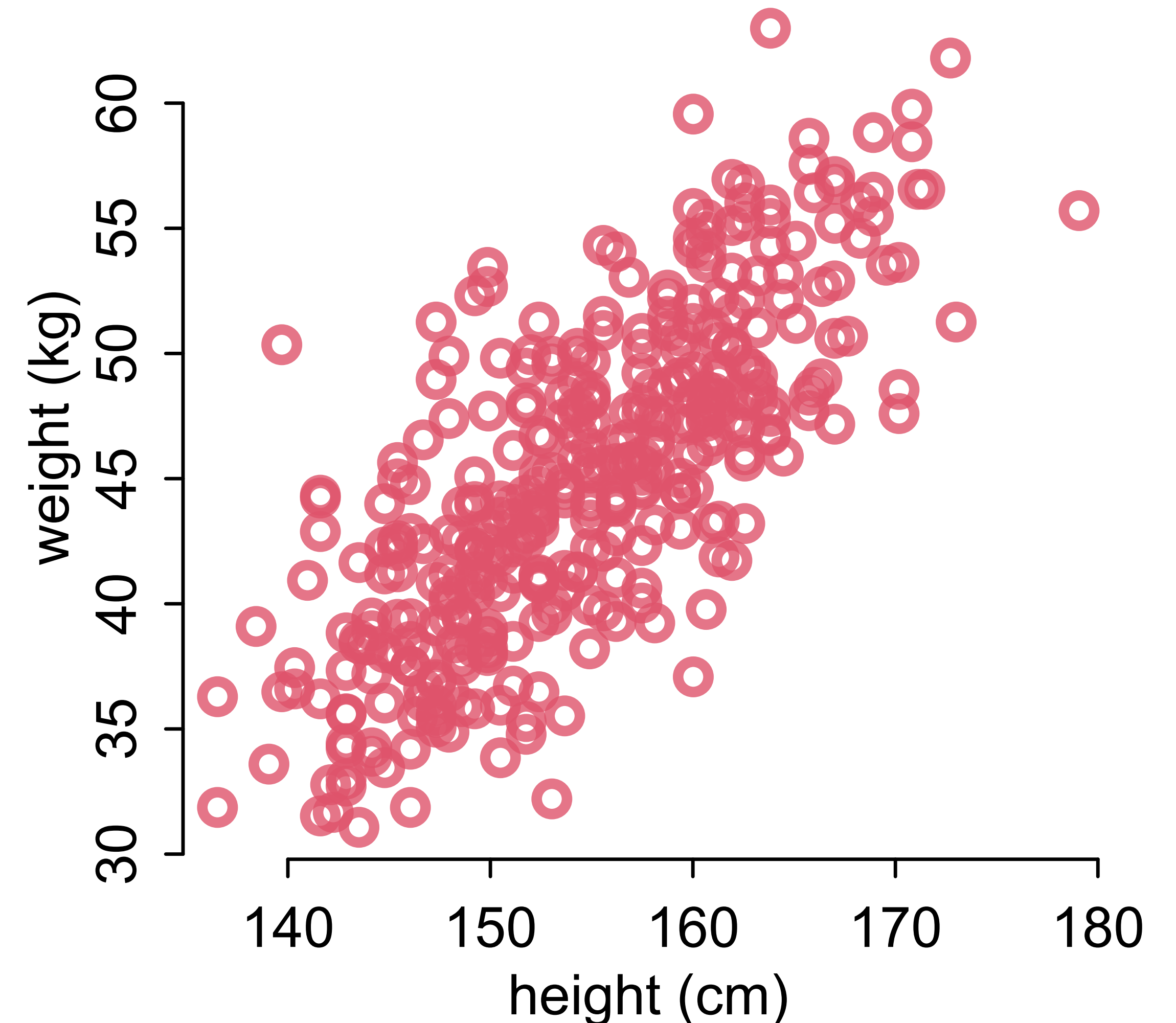
(1) Question/goal/estimand

(2) Scientific model

(3) Statistical model(s)

(4) Validate model

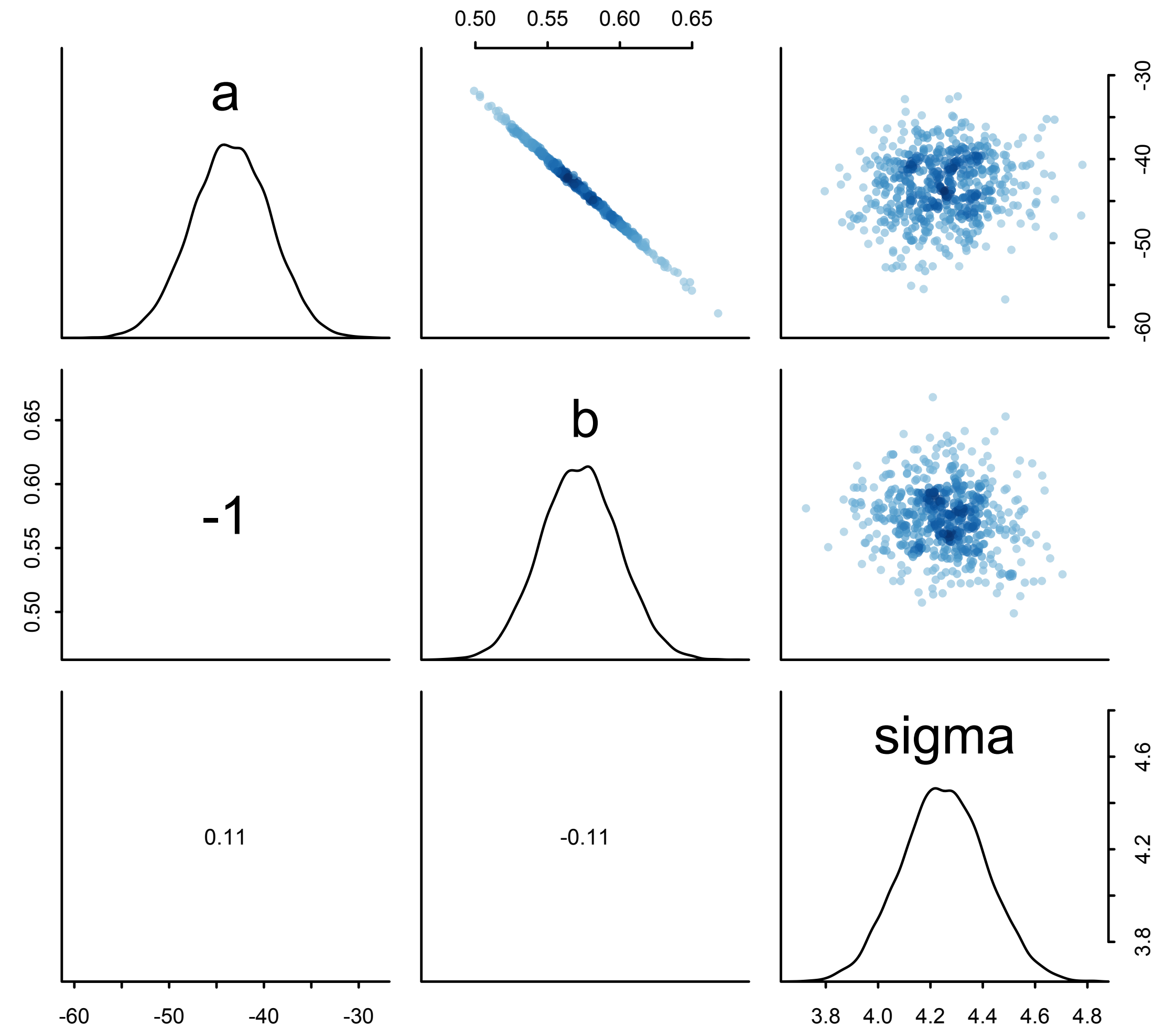
(5) Analyze data



Analyze the data

```
dat <- list(W=d2$weight,H=d2$height)
m3.2 <- quap(
  alist(
    W ~ dnorm(mu,sigma),
    mu <- a + b*H,
    a ~ dnorm(0,10),
    b ~ dunif(0,1),
    sigma ~ dunif(0,10)
  ) , data=dat )
precis( m3.2 )
```

	mean	sd	5.5%	94.5%
a	-43.38	4.17	-50.04	-36.71
b	0.57	0.03	0.53	0.61
sigma	4.25	0.16	3.99	4.51



Obey The Law

First Law of Statistical Interpretation:

The **parameters are not independent** of one another and cannot always be independently interpreted

Instead:

Push out **posterior predictions** and describe/interpret those

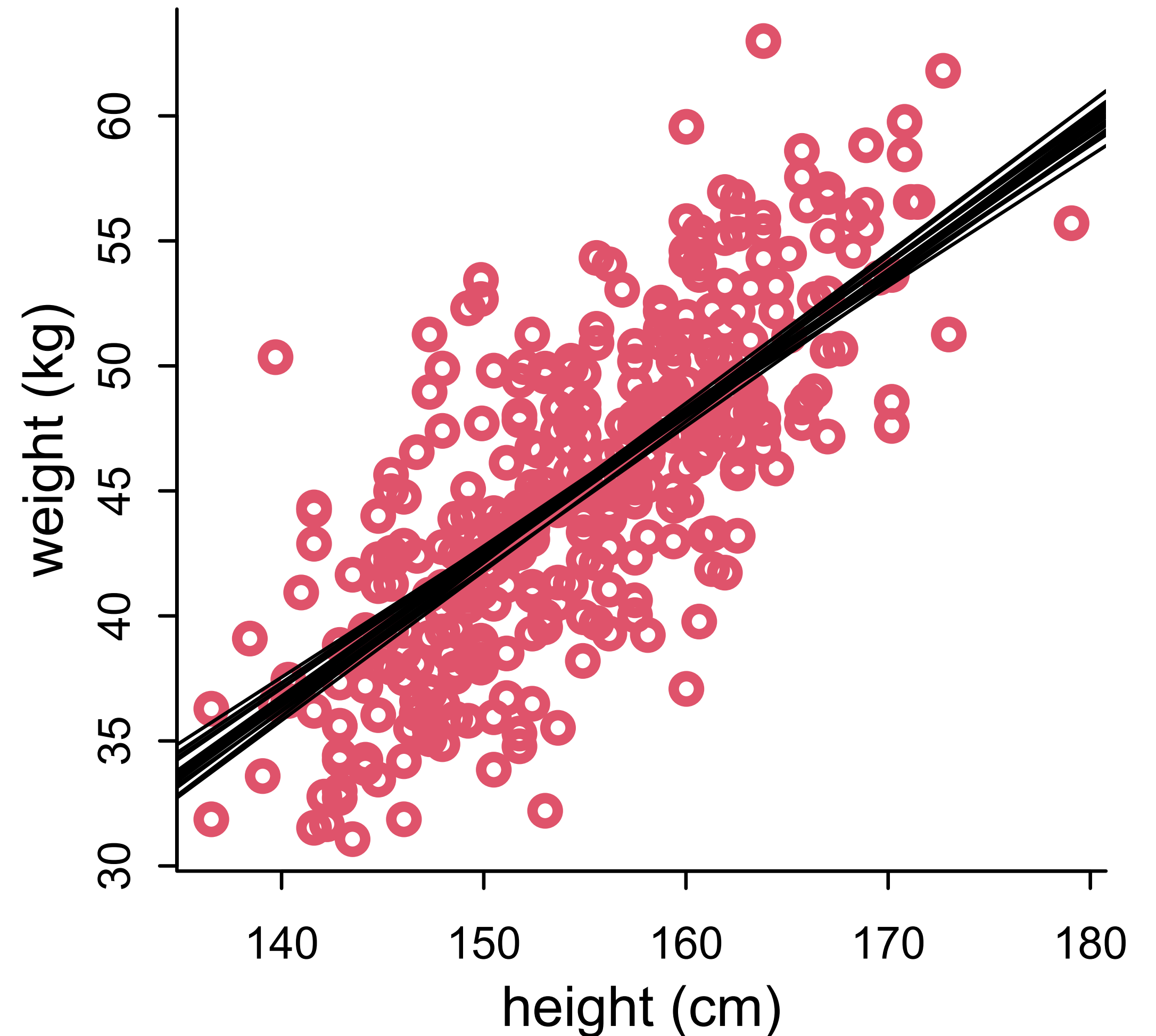
```
> precis(m_adults)
      mean    sd  5.5% 94.5%
a    45.00  0.23 44.64 45.36
b     0.63  0.03  0.58  0.68
sigma 4.23  0.16  3.97  4.48
>
```

```
> post <- extract.samples(m_adults)
> head(post)
      a          b      sigma
1 45.14733 0.7045790 4.380254
2 44.97759 0.6461353 4.372925
3 44.94856 0.6537192 4.111149
4 44.85016 0.6597310 4.379347
5 44.75898 0.6532690 4.200026
6 44.91711 0.6090434 4.105432
>
```

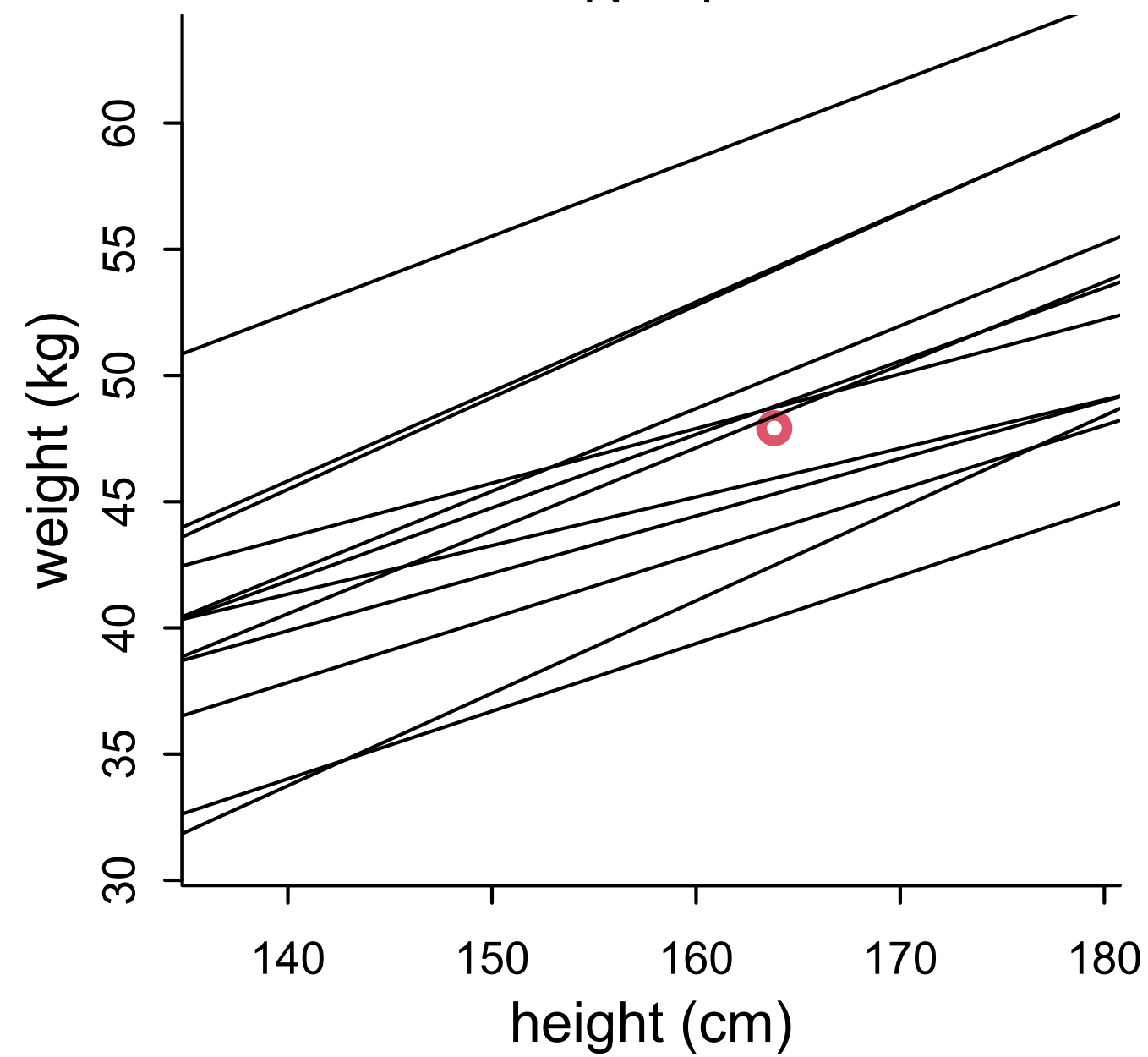
Posterior predictive distribution

The posterior is full of lines

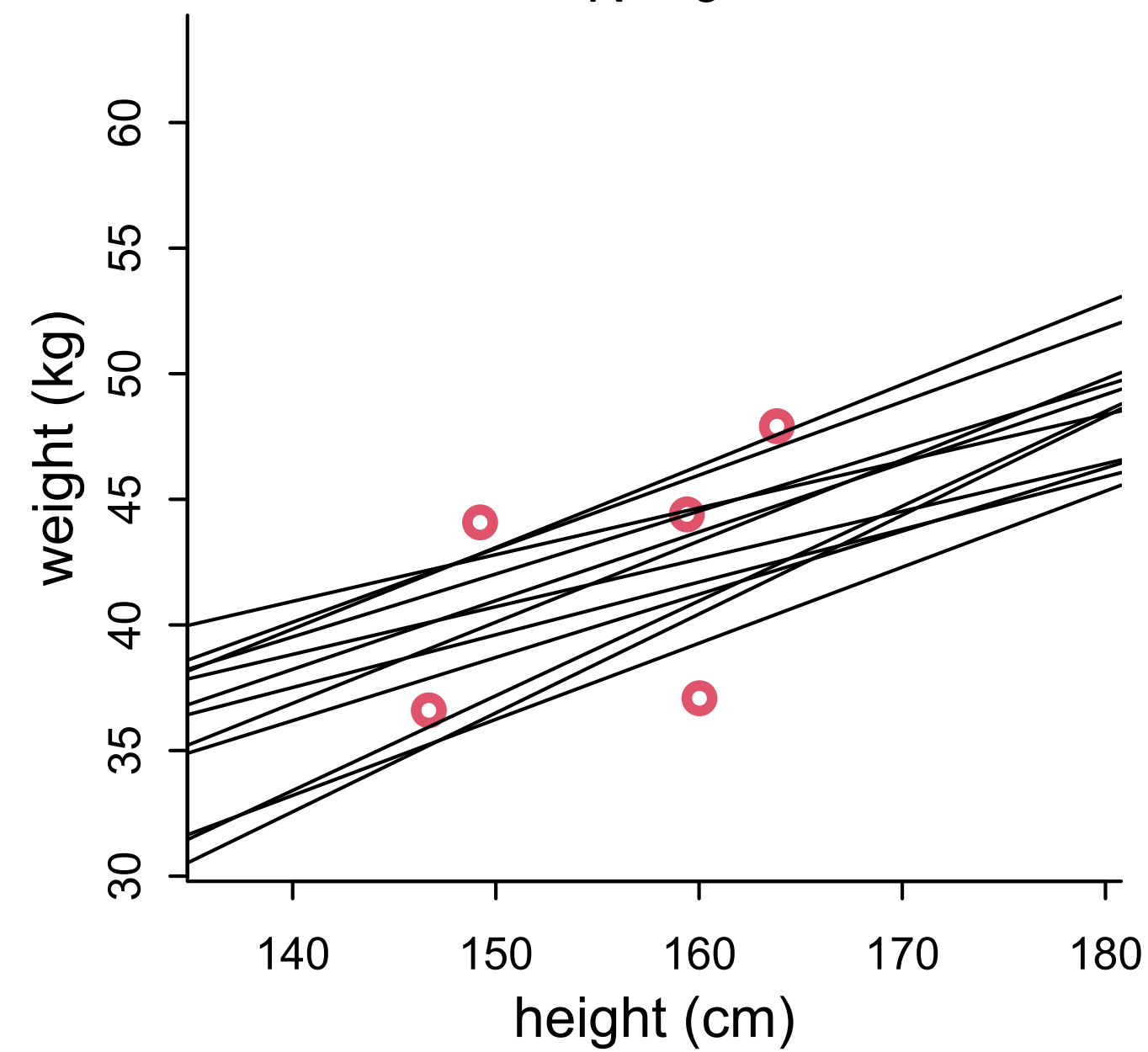
```
post <- extract.samples(m3.2)
plot( d2$height , d2$weight , col=2 , lwd=3 ,
      xlab="height (cm)" , ylab="weight (kg)" )
for ( j in 1:20 )
  abline( a=post$a[j] , b=post$b[j] , lwd=1 )
```



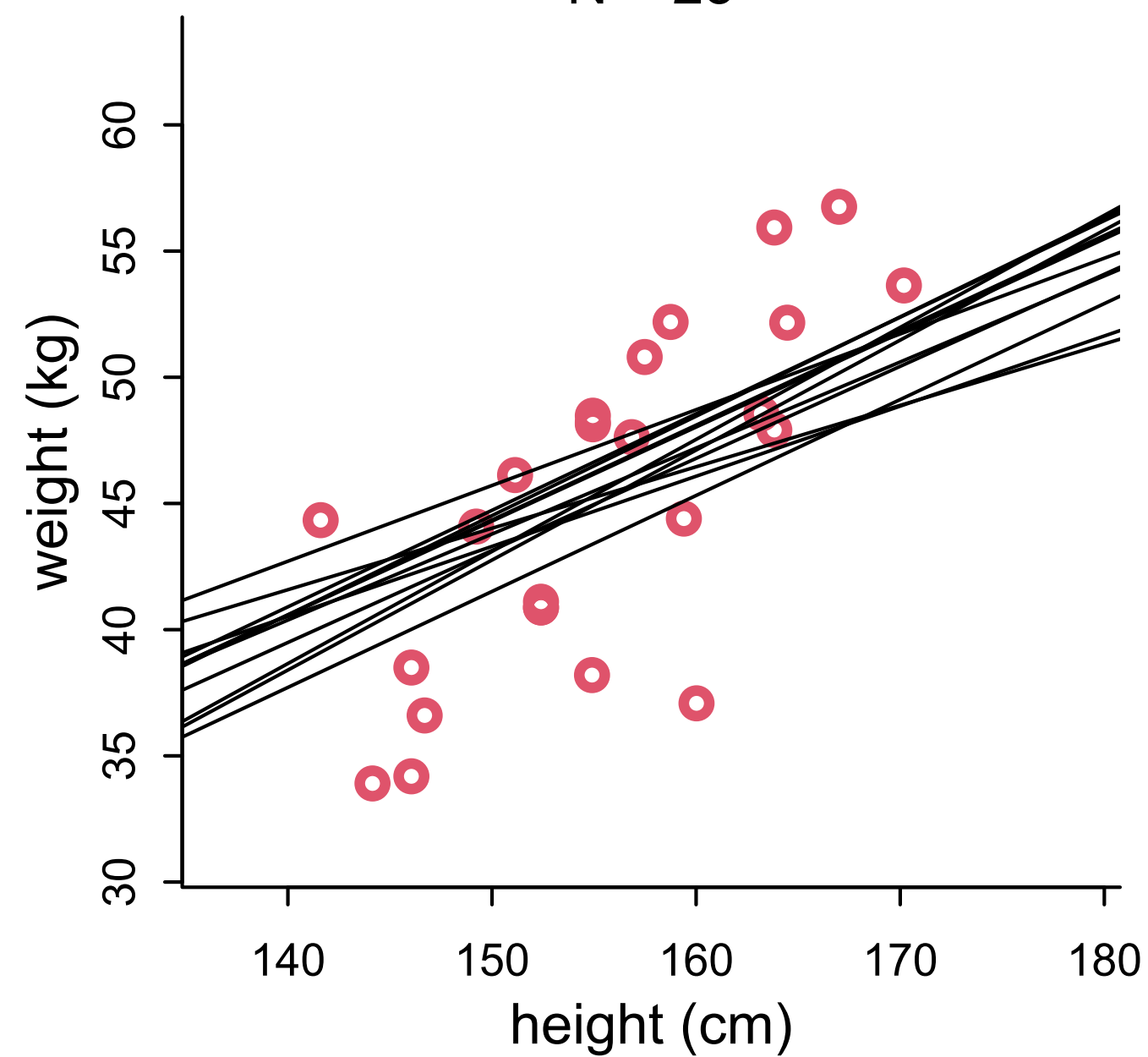
N = 1



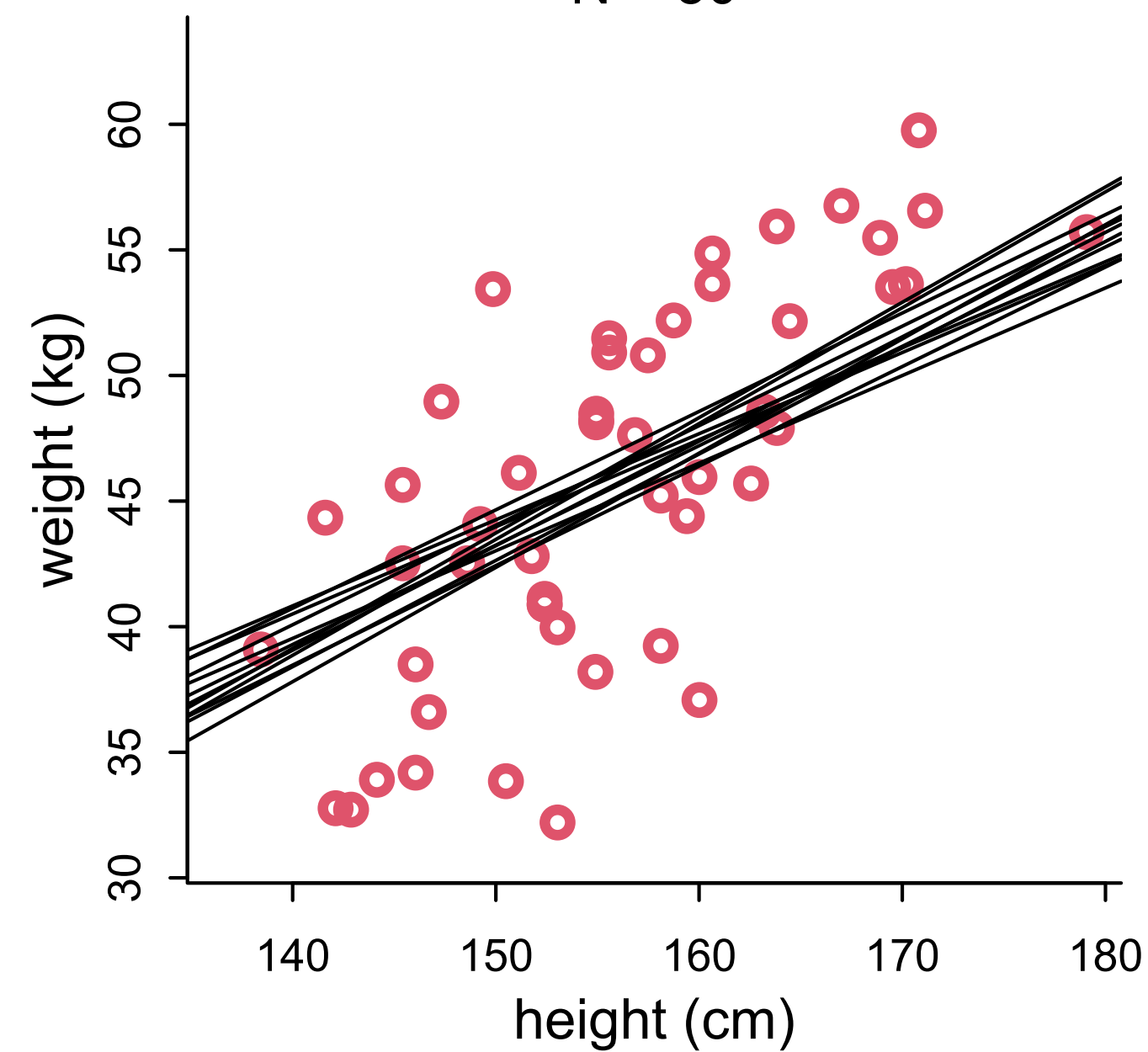
N = 5



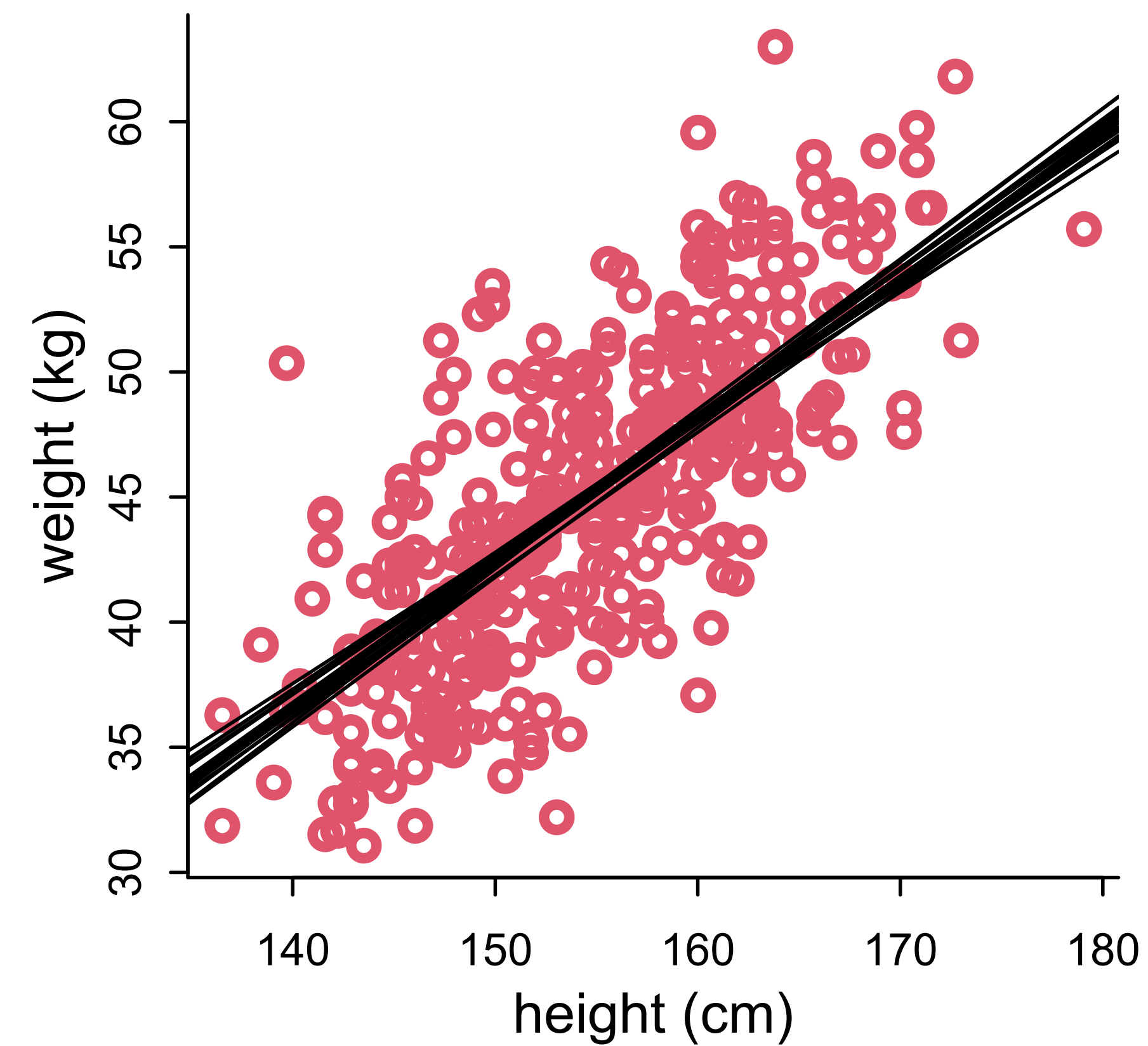
N = 25



N = 50



N = 352



Posterior predictive distribution

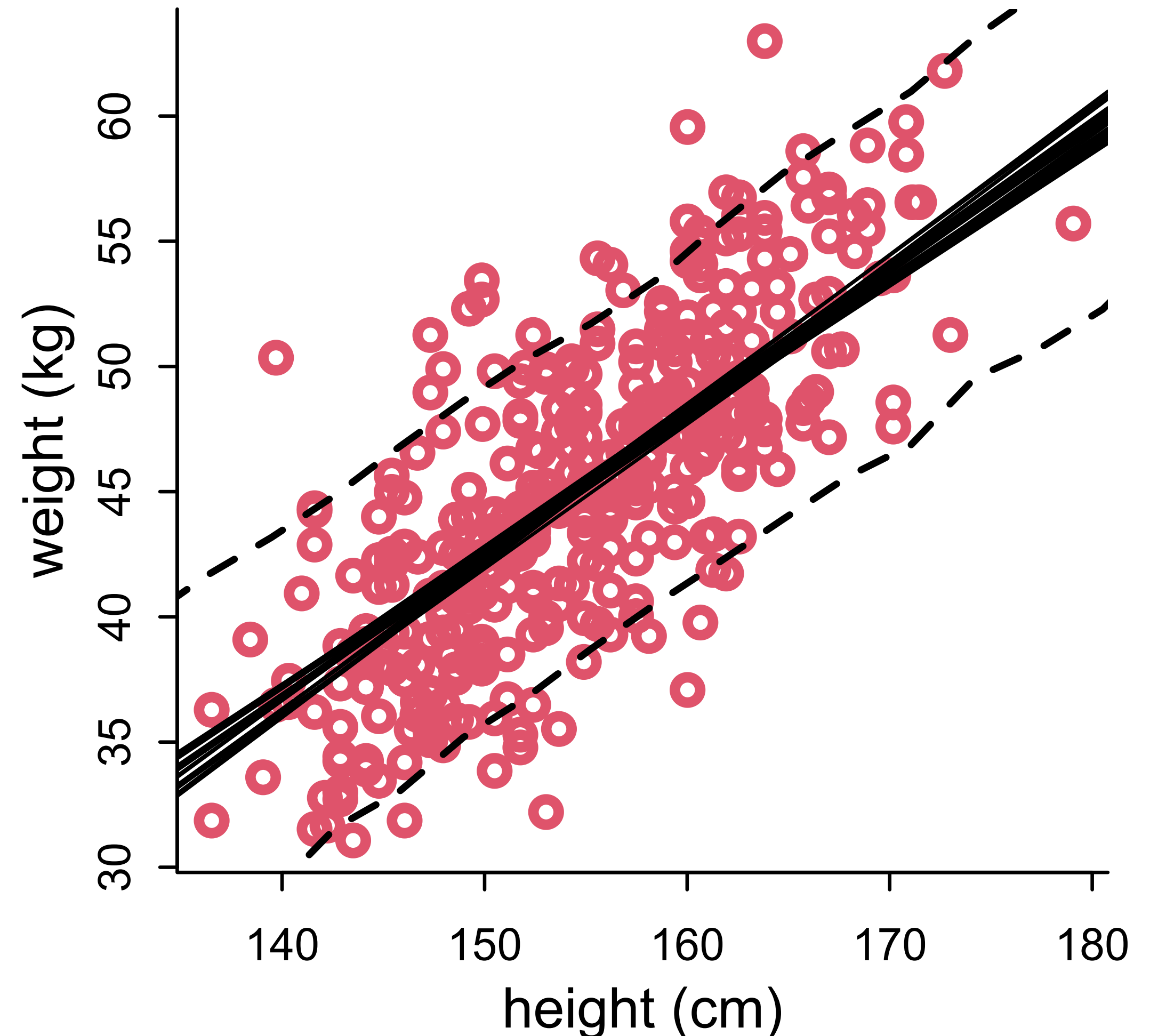
The posterior is full of lines

```
post <- extract.samples(m3.2)
plot( d2$height , d2$weight , col=2 , lwd=3 ,
      xlab="height (cm)" , ylab="weight (kg)" )
for ( j in 1:20 )
  abline( a=post$a[j] , b=post$b[j] , lwd=1 )
```

The posterior is full of people

```
height_seq <- seq(130,190,len=20)
W_postpred <- sim( m3.2 ,
  data=list(H=height_seq) )
W_PI <- apply( W_postpred , 2 , PI )
lines( height_seq , W_PI[1,] , lty=2 , lwd=2 )
lines( height_seq , W_PI[2,] , lty=2 , lwd=2 )
```

***PI = percentile interval function**



Generative model

How does **height** influence **weight**?



$$W = f(H, U)$$

“Weight is some function of height & unmeasured stuff”

Statistical model

How does average **weight** change with **height**?

$$W_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta H_i$$

$$\alpha \sim \text{Normal}(0, 10)$$

$$\beta \sim \text{Uniform}(0, 1)$$

$$\sigma \sim \text{Uniform}(0, 10)$$